

DESIGN AND IMPLEMENTATION OF A SECURE ENTERPRISE NETWORK INFRASTRUCTURE

Using GNS3, pfSense, Dynamic Routing, DMZ, VPN, and Network Security Services

CHAPTER 1 — INTRODUCTION:

1.1 Project Overview:

This project presents the design and implementation of a secure enterprise network infrastructure using GNS3, pfSense, Cisco routers, Ubuntu Server, Kali Linux, and OpenVPN.

The main purpose of the project is to build a realistic network environment where different organizations, servers, and users can communicate with each other while security controls are applied to protect the internal network.

The network contains an ISP/core routing environment, company networks, a dedicated DMZ, a firewall, and an Ubuntu-based server providing multiple network services. Routing protocols such as OSPF and BGP are used to demonstrate communication between different autonomous systems and internal networks.

The project also implements a pfSense firewall as the central security device. It controls traffic between the Internet, company networks, and the DMZ. Network Address Translation (NAT), firewall rules, port forwarding, DNS forwarding, and OpenVPN are configured to provide controlled and secure access to network resources.

The Ubuntu Server is used to provide several important services, including:

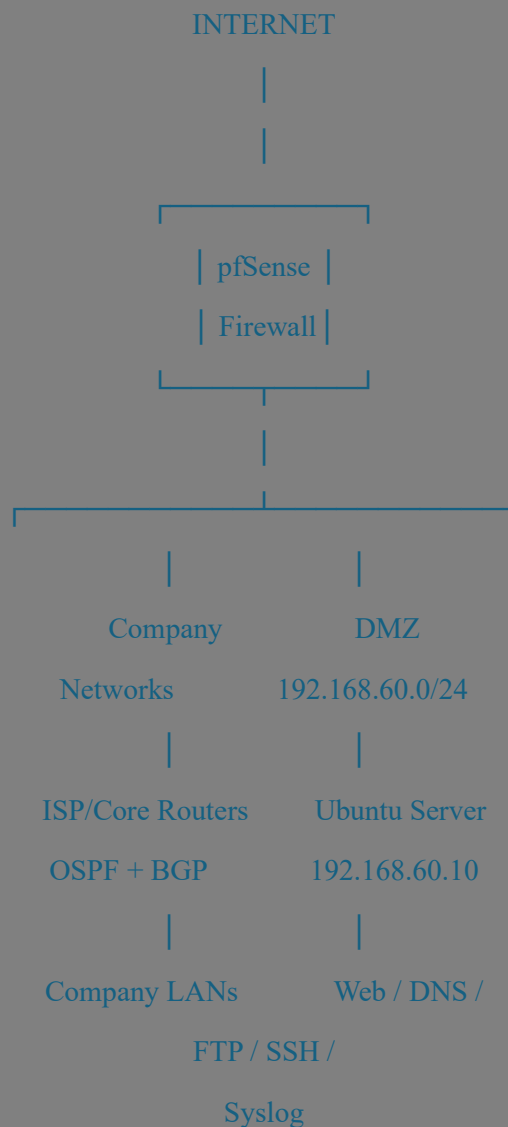
- Web server using Apache HTTP/HTTPS
- DNS server using BIND9
- FTP server using vsftpd
- SSH remote administration
- Centralized logging using rsyslog
- DMZ-hosted web services

The project also includes a DMZ (Demilitarized Zone) to isolate publicly accessible services from the internal company network. Security testing is performed using Kali Linux and tools such as Nmap, Netcat, curl, ping, and dig to verify that the configured services are accessible while unauthorized traffic is blocked.

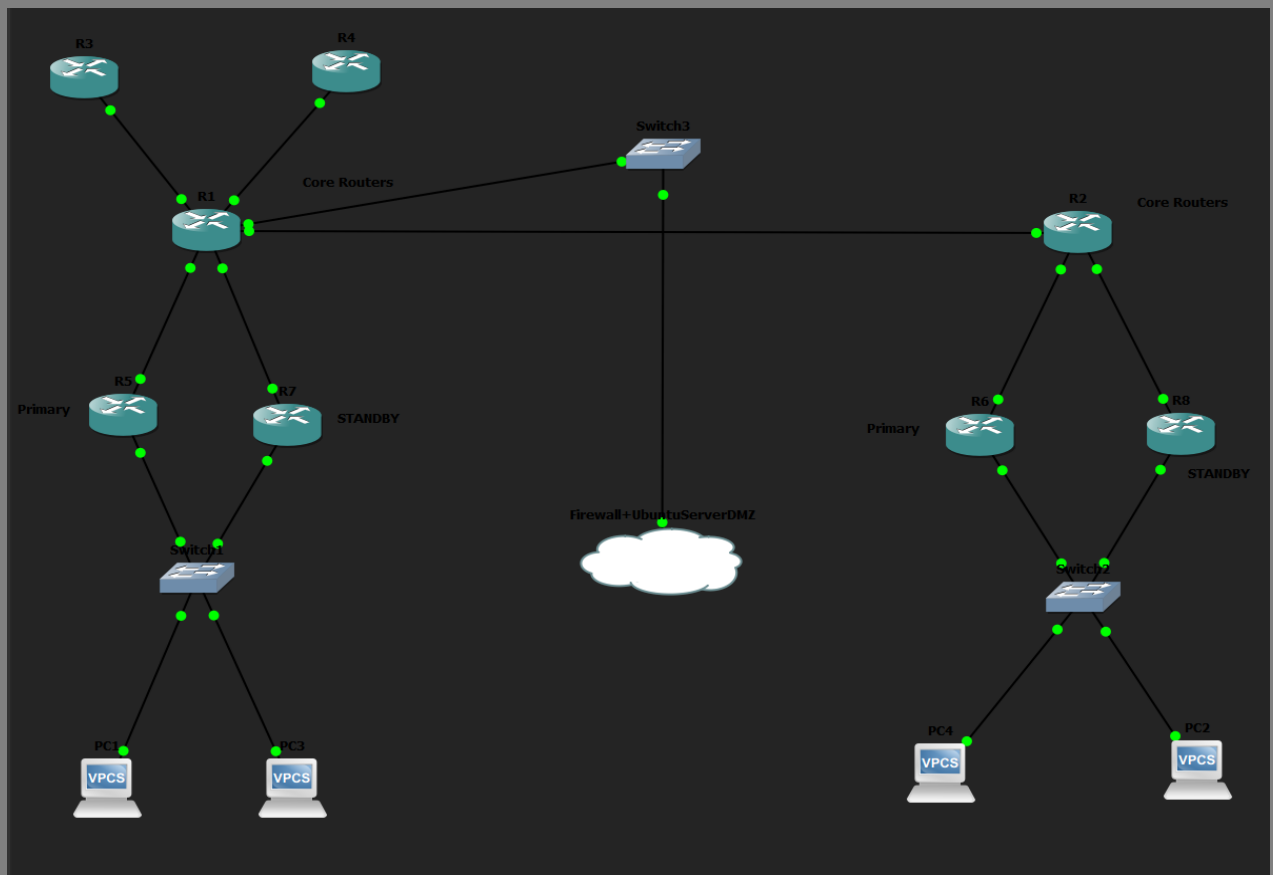
1.2 Network Architecture:

The overall architecture is designed around the principle of network segmentation and controlled communication.

At the edge of the network, pfSense acts as the primary firewall and security gateway. Behind pfSense, different network segments are connected through the routing infrastructure. The company networks are connected through the ISP/core routers, while the Ubuntu server is connected to the DMZ.



Network Topology:



1.3 Purpose of the Project:

The project is designed to demonstrate how networking and cybersecurity technologies can be combined to build a functional and secure enterprise environment.

Instead of creating a network where every device can communicate freely, the project applies security controls based on the purpose of each network segment.

For example, public-facing services are placed in the DMZ rather than directly inside the company network. pfSense controls which services can be accessed from external networks, while routing protocols provide connectivity between different parts of the simulated enterprise infrastructure.

This approach provides both network functionality and security, which are essential requirements in modern enterprise environments.

1.4 Technologies Used:

The project uses the following major technologies:

Technology	Purpose
GNS3	Network simulation and topology
pfSense	Firewall, NAT, routing/security gateway and VPN
Cisco Routers	ISP/core routing
OSPF	Internal dynamic routing
BGP	Inter-network/ISP routing
Ubuntu Server	Network services and DMZ server
Apache	HTTP/HTTPS web service
BIND9	DNS service
vsftpd	FTP service
OpenSSH	Secure remote administration
rsyslog	Centralized logging
OpenVPN	Remote-access VPN
Kali Linux	Security testing and verification
Nmap	Port and service scanning
Netcat	Connectivity testing
curl	HTTP/HTTPS testing
dig	DNS testing

1.5 Security Approach:

Security is an important part of the project. The network does not simply provide connectivity; it also controls and monitors communication between different network zones.

The main security mechanisms implemented include:

- Firewall rules to control traffic.
- NAT and port forwarding for controlled external access.
- DMZ segmentation for public-facing services.
- Restricted service ports instead of allowing unrestricted access.
- OpenVPN for secure remote access.
- DNS domain forwarding through pfSense.
- Centralized Syslog logging for network events.
- Port scanning and connectivity testing using Kali Linux.
- DMZ isolation testing to verify that internal networks are protected.
- Service-specific firewall rules allowing only required traffic.

The project therefore demonstrates the principle of allowing required communication while blocking unnecessary or unauthorized traffic.

1.6 Scope of the Project:

The scope of this project includes the design, configuration, testing, and security verification of the simulated enterprise network.

The project covers:

1. Enterprise network topology design.
2. OSPF and BGP routing.
3. pfSense firewall deployment.
4. NAT and port forwarding.
5. DMZ implementation.
6. Ubuntu Server deployment.

7. HTTP and HTTPS services.
8. DNS implementation using BIND9.
9. FTP implementation using vsftpd.
- 10.SSH remote administration.
- 11.Centralized Syslog.
- 12.OpenVPN remote access.
- 13.Firewall and security rule testing.
- 14.Network and service verification using Kali Linux.
- 15.DMZ isolation and port-security testing.

The project does not currently implement VLAN segmentation because the available GNS3 environment does not contain an IOS Layer-2 switch image. VLAN implementation can be considered a future enhancement.

1.7 Chapter Summary:

This chapter introduced the enterprise network security project and described its overall architecture, purpose, technologies, scope, and security approach.

The project combines routing, firewalling, network segmentation, server services, VPN, monitoring, and security testing into one practical environment. The following chapters will explain each component in detail, including its configuration, purpose, testing procedure, and results.

CHAPTER 2 — PROJECT OBJECTIVES AND REQUIREMENTS

2.1 Project Objectives

The primary objective of this project is to design and implement a secure, functional, and realistic enterprise network infrastructure using network simulation and security technologies.

The project focuses on combining networking and cybersecurity concepts into a single practical environment rather than implementing each technology separately.

The main objectives are:

1. Design an Enterprise Network Topology:

To design a realistic enterprise network consisting of:

- ISP/core routers
- Company networks
- Data center/DMZ
- Firewall
- Network services
- Remote-access VPN

The topology is implemented and tested using GNS3.

2. Implement Dynamic Routing:

To implement dynamic routing protocols for reliable communication between different networks.

The project uses:

- **OSPF** for internal routing
- **BGP** for communication between different autonomous systems
- **HSRP** for gateway redundancy/high availability were implemented

The purpose is to demonstrate how enterprise networks maintain connectivity and route traffic dynamically.

3. Implement a Secure Firewall:

To deploy pfSense as the central firewall and security gateway.

The firewall is responsible for:

- Controlling inbound and outbound traffic
- Network segmentation
- NAT
- Port forwarding
- DMZ protection
- VPN access
- Logging and monitoring

4. Implement a DMZ:

To create a dedicated Demilitarized Zone (DMZ) for publicly accessible services.

The DMZ prevents public-facing services from being placed directly inside the internal company network.

The Ubuntu server in the DMZ provides services such as:

- Web server
- HTTPS
- FTP
- SSH
- Syslog

5. Implement Network Services:

To configure and test common enterprise network services on Ubuntu Server, including:

- Apache HTTP/HTTPS
- BIND9 DNS
- vsftpd FTP

- OpenSSH
- rsyslog

These services demonstrate how a centralized server can provide multiple network functions.

6. Implement Secure Remote Access:

To configure OpenVPN on pfSense for secure remote access to the enterprise network.

The VPN provides an encrypted communication channel between authorized remote users and the internal network.

7. Apply Network Security Controls:

To implement security rules based on the principle of least privilege.

Only required services and ports are allowed between network zones, while unnecessary or unauthorized traffic is blocked.

8. Perform Security Testing:

To verify the security and functionality of the network using tools such as:

- Nmap
- Netcat
- curl
- dig
- ping
- Linux logging utilities

Testing is performed from Kali Linux to verify service availability, firewall behavior, DMZ isolation, and network connectivity.

9. Implement Network Monitoring and Logging:

To configure centralized logging using rsyslog so that network events and messages can be collected and reviewed.

This provides visibility into network activity and helps with troubleshooting and security monitoring.

2.2 Security Requirements:

The project follows several basic network-security requirements.

Controlled Access

Only required services should be accessible from external networks.

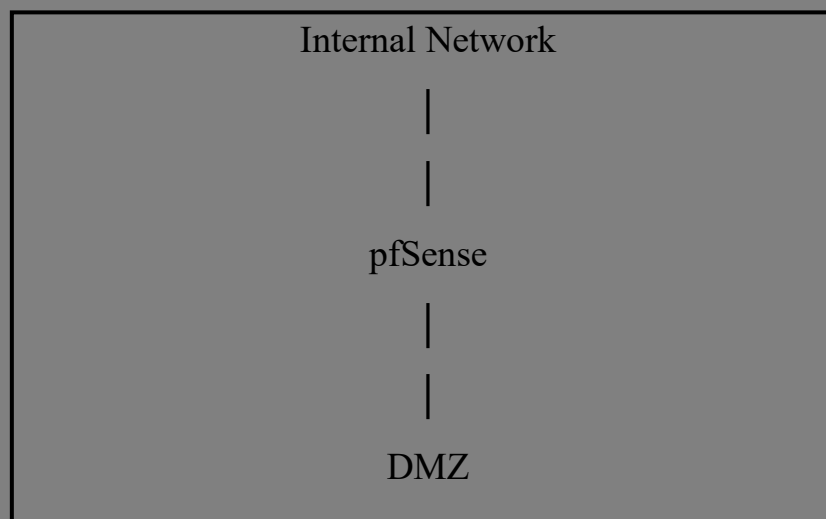
For example:

WAN → HTTP	Allowed
WAN → HTTPS	Allowed
WAN → FTP	Blocked
WAN → SSH	Blocked
WAN → Syslog	Blocked

unless a specific rule is intentionally created for a controlled source.

Network Segmentation:

The internal network and DMZ should remain logically separated.



DMZ Isolation:

Traffic originating from the DMZ toward protected internal networks should be restricted.

This was verified during the project using controlled routing and firewall testing.

Logging:

Important network and service activity should be logged for troubleshooting and security analysis.

Secure Administration:

SSH should be used instead of insecure remote-administration protocols.

VPN access should use encrypted communication through OpenVPN.

2.4 Testing Requirements:

The completed infrastructure must be tested from both functionality and security perspectives.

Testing includes:

Routing connectivity	→ Ping / routing tables
BGP/OSPF	→ Routing verification
NAT	→ External connectivity
DMZ	→ Port/service testing
HTTP/HTTPS	→ curl/browser
DNS	→ dig
FTP	→ FTP client/Nmap
SSH	→ SSH client/Nmap
Syslog	→ logger/tail
VPN	→ OpenVPN client
Firewall	→ Allowed/blocked traffic
DMZ isolation	→ Controlled routing tests
Port security	→ Nmap

2.6 Project Constraints:

During implementation, some limitations were encountered.

The current GNS3 environment does not contain an IOS Layer-2 switch image, therefore VLAN implementation could not be added to the final topology.

Additionally, the Ubuntu DMZ server uses two network interfaces. This means that traffic toward the internal network can potentially bypass the pfSense DMZ interface through the server's internal interface. This limitation was identified during testing and is documented as an architectural consideration.

A future improvement would be to use a dedicated single-homed DMZ server, and an appropriate Layer-2 switch to implement complete VLAN-based segmentation.

CHAPTER 3 — NETWORK ARCHITECTURE AND IP ADDRESSING

3.1 Network Architecture Overview:

The proposed network architecture represents a simulated enterprise environment consisting of an Internet/ISP network, core routing infrastructure, company networks, a pfSense security gateway, and a dedicated DMZ.

The network was implemented using GNS3 and virtual machines. Cisco routers provide the routing infrastructure, while pfSense provides firewalling, NAT, DMZ security, and VPN functionality.

The architecture is divided into multiple logical network segments to improve security, scalability, and manageability.

3.3 Major Network Components:

The project contains the following major components.

3.3.1 pfSense Firewall

pfSense is deployed as the primary security gateway.

Its main functions include:

- Firewall filtering
- NAT

- Port forwarding
- DMZ protection
- DNS forwarding
- OpenVPN
- Traffic logging
- Network segmentation

The firewall has multiple interfaces connecting different network zones.

3.3.2 ISP/Core Routing Infrastructure:

Cisco routers are used to simulate an ISP/core network.

The routing infrastructure uses:

- **BGP** for autonomous-system-level routing
- **OSPF** for internal dynamic routing
- **HSRP** for gateway redundancy were implemented

This provides a realistic representation of enterprise and ISP routing.

3.3.3 Company Networks:

The company networks represent internal enterprise users and resources.

These networks are connected through the routing infrastructure and are protected from external networks by pfSense.

The company networks use private IP addressing.

3.3.4 DMZ:

A dedicated DMZ network is used for services that require controlled access from external networks.

The DMZ network is:

192.168.60.0/24

The Ubuntu DMZ server uses:

192.168.60.10

The DMZ contains services including:

- HTTP
- HTTPS
- FTP
- SSH
- Syslog

Traffic to these services is controlled using pfSense firewall rules.

3.4 Network Interfaces and Addressing:

The main Ubuntu server interfaces are configured as follows:

Interface	IP Address	Network	Purpose
ens33	192.168.50.21/24	192.168.60.0/24	Internal/service network
ens37	192.168.60.10/24	192.168.60.0/24	DMZ

The Ubuntu server therefore provides services to both the internal/service environment and the DMZ.

```
haider@linux-server:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:31:20:1b brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.50.21/24 brd 192.168.50.255 scope global ens33
        valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fe31:201b/64 scope link
        valid_lft forever preferred_lft forever
3: ens37: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:31:20:25 brd ff:ff:ff:ff:ff:ff
    altname enp2s5
    inet 192.168.60.10/24 brd 192.168.60.255 scope global ens37
        valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fe31:2025/64 scope link
        valid_lft forever preferred_lft forever
haider@linux-server:~$
```

3.5 Main IP Addressing Scheme:

The following table summarizes the major addresses used in the project

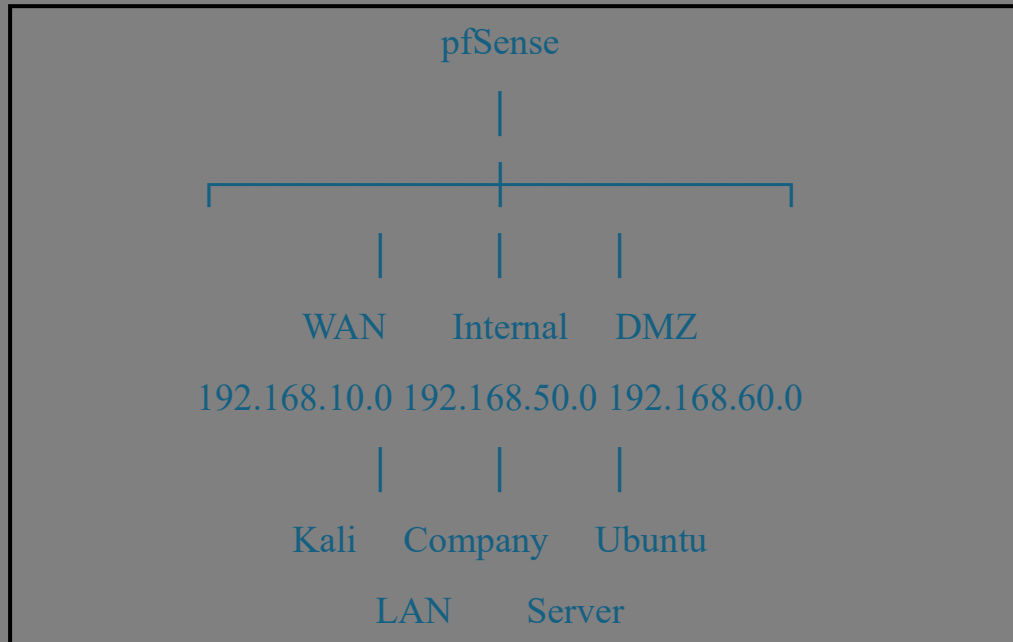
Device/Interface	IP Address	Role
pfSense WAN	192.168.10.10	External/WAN gateway
Kali Linux	192.168.10.13	Security testing
pfSense Internal/OPT1	192.168.50.1	Internal gateway
Ubuntu ens33	192.168.50.21	Internal/service interface
pfSense DMZ/OPT2	192.168.60.1	DMZ gateway
Ubuntu ens37	192.168.60.10	DMZ server
DMZ Network	192.168.60.0/24	Public-service network

Note: Other router addresses are part of the GNS3 ISP/company routing topology and are documented according to their respective routing sections.

3.6 Network Segmentation:

Network segmentation is one of the main security principles used in this project.

The architecture separates:



The firewall controls communication between these zones.

For example, external clients can access specifically permitted DMZ services, while unnecessary access to internal networks is blocked.

3.7 DMZ Architecture:

The DMZ is designed to host services that may need external access without placing those services directly inside the protected internal network.

External Client



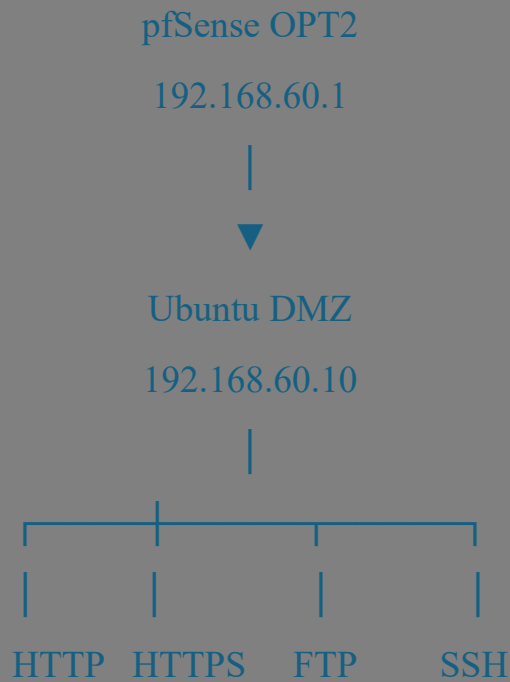
pfSense WAN

192.168.10.10



Firewall/NAT





The DMZ security rules allow only required services.

Interfaces			
 WAN	↑	1000baseT <full-duplex>	192.168.10.10
 LAN	↑	1000baseT <full-duplex>	192.168.75.2
 OPT1	↑	1000baseT <full-duplex>	192.168.50.1
 OPT2	↑	1000baseT <full-duplex>	192.168.60.1

3.8 Internet and WAN Connectivity:

The WAN network connects pfSense to the simulated external/ISP environment.

Kali Linux is located on the WAN-side test network:

Kali:

192.168.10.13

```
(kali㉿kali)-[~]
$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.10.13 netmask 255.255.255.0 broadcast 192.168.10.255
    inet6 fe80::20c:29ff:fe38:e14f prefixlen 64 scopeid 0x20<link>
    ether 00:0c:29:38:e1:4f txqueuelen 1000 (Ethernet)
    RX packets 1163 bytes 103329 (100.9 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 352 bytes 31185 (30.4 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

pfSense provides the WAN gateway:

192.168.10.10

Interfaces			
 WAN	↑	1000baseT <full-duplex>	192.168.10.10
 LAN	↑	1000baseT <full-duplex>	192.168.75.2
 OPT1	↑	1000baseT <full-duplex>	192.168.50.1
 OPT2	↑	1000baseT <full-duplex>	192.168.60.1

Controlled firewall rules are used to allow specific services from the WAN toward the DMZ.

For example:

WAN → TCP 80 → DMZ Web Server

WAN → TCP 443 → DMZ Web Server

Other unnecessary services are blocked.

3.9 Routing Architecture:

Routing is divided into internal and external routing functions.

OSPF:

OSPF is used within the routing infrastructure to dynamically exchange routes between routers.

BGP:

BGP is used to represent routing between different autonomous systems and ISP/company networks.

HSRP:

HSRP is used where gateway redundancy is implemented. It provides a virtual gateway address so that another router can take over if the active router becomes unavailable.

3.10 Design Considerations:

Several design decisions were made to preserve the stability of the project.

The existing 192.168.50.0/24 network was retained because it is already integrated with the GNS3 company network.

The Ubuntu server was connected to the DMZ through its ens37 interface:

ens37 → 192.168.60.10

This allowed the DMZ services to be exposed through pfSense without changing the existing company addressing structure.

A limitation of the current architecture is that Ubuntu is dual homed, meaning it has both an internal and DMZ interface. A production environment would normally use a dedicated, appropriately segmented DMZ server to prevent possible bypass of firewall controls through another interface.

CHAPTER 4 — ROUTING AND HIGH AVAILABILITY

4.1 Introduction:

Routing is a fundamental component of the enterprise network because it enables communication between different networks and ensures that traffic reaches its intended destination.

In this project, multiple routing technologies are used to represent a realistic enterprise and ISP environment. OSPF is used for internal dynamic routing, while BGP is used to exchange routes between different Autonomous Systems (AS). HSRP is implemented to provide first-hop gateway redundancy and maintain connectivity when an active gateway router becomes unavailable.

The combination of OSPF, BGP, and HSRP provides:

- Dynamic route learning
- Inter-network route exchange
- Redundant gateway connectivity
- Primary and backup paths
- Improved network availability
- Automatic route convergence after failures

4.2 OSPF — Open Shortest Path First:

OSPF is the internal routing protocol used within the enterprise/core routing environment.

It is a link-state routing protocol that allows routers within the same routing domain to exchange information about their connected networks and calculate the best available path.

The project uses OSPF Area 0 as the main backbone area.

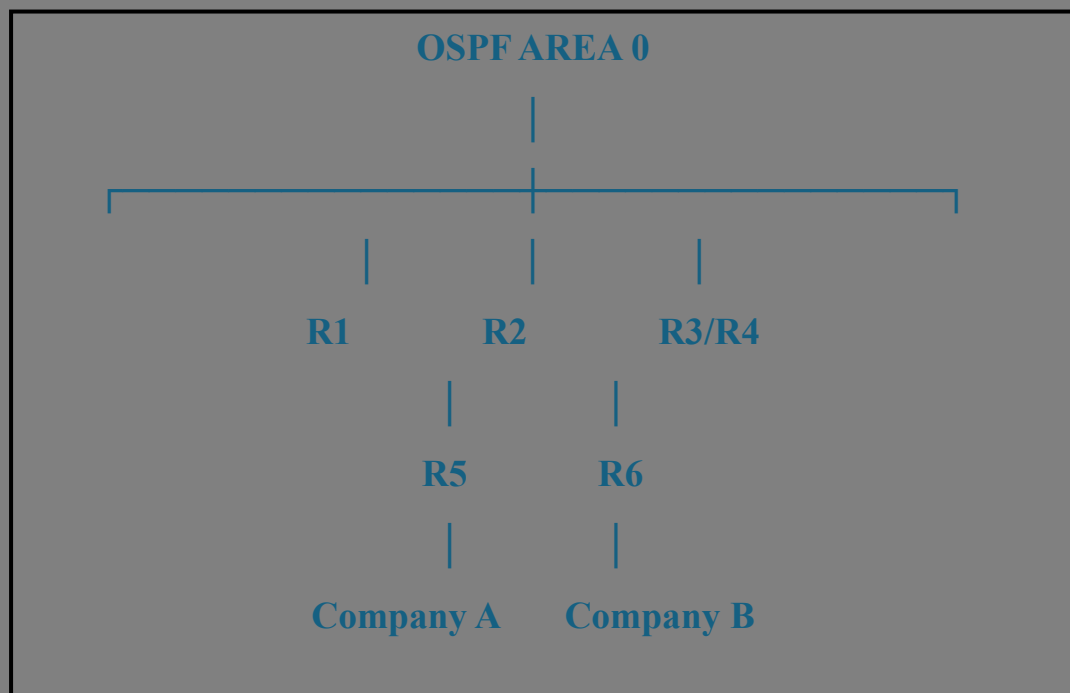
OSPF Objectives in the Project

OSPF is used to:

- Dynamically exchange internal routes
- Advertise router loopback addresses
- Provide automatic route learning
- Reduce the need for manually configured routes
- Support connectivity between the core and company networks

The routers participate in OSPF using their loopback addresses as router IDs.

Example OSPF Design:



4.3 OSPF Router IDs:

Loopback interfaces are used as stable OSPF router IDs.

Examples from the project include:

Router	Loopback / Router ID
R1	1.1.1.1
R2	2.2.2.2
R3	3.3.3.3
R4	4.4.4.4
R5	5.5.5.5
R6	6.6.6.6
R7	7.7.7.7
R8	8.8.8.8

These loopbacks provide stable router identification even if an individual physical interface changes state.

```
R1#show run | section router ospf
router ospf 1
  router-id 1.1.1.1
  log-adjacency-changes
  network 1.1.1.1 0.0.0.0 area 0
  network 10.0.0.0 0.0.0.255 area 0
  network 172.16.100.0 0.0.0.255 area 0
  network 192.168.10.0 0.0.0.255 area 0
  network 192.168.50.0 0.0.0.255 area 0
R1#
```

4.4 OSPF Route Learning:

After OSPF adjacency is established, routers dynamically learn remote networks.

For example, R1 learns networks located behind other routers without requiring a separate static route for every destination.

OSPF routes appear in the routing table with the code: O

This demonstrates that the route was learned dynamically through OSPF.

OSPF Neighbor:

```
R1#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
7.7.7.7	1	FULL/DR	00:00:36	10.0.0.22	Ethernet2/0
5.5.5.5	1	FULL/DR	00:00:33	10.0.0.14	Ethernet1/2
4.4.4.4	1	FULL/DR	00:00:37	10.0.0.10	Ethernet1/1
3.3.3.3	1	FULL/DR	00:00:34	10.0.0.6	Ethernet1/0
2.2.2.2	1	FULL/DR	00:00:36	10.0.0.2	FastEthernet0/0

```
R1#
```

IP OSPF:

```
R1#show ip ospf
Routing Process "ospf 1" with ID 1.1.1.1
Start time: 00:00:33.496, Time elapsed: 00:39:57.268
Supports only single TOS(TOS0) routes
Supports opaque LSA
Supports Link-local Signaling (LLS)
Supports area transit capability
Router is not originating router-LSAs with maximum metric
Initial SPF schedule delay 5000 msecs
Minimum hold time between two consecutive SPF's 10000 msecs
Maximum wait time between two consecutive SPF's 10000 msecs
Incremental-SPF disabled
Minimum LSA interval 5 secs
Minimum LSA arrival 1000 msecs
LSA group pacing timer 240 secs
Interface flood pacing timer 33 msecs
Retransmission pacing timer 66 msecs
Number of external LSA 1. Checksum Sum 0x007925
Number of opaque AS LSA 0. Checksum Sum 0x000000
Number of DCbitless external and opaque AS LSA 0
Number of DoNotAge external and opaque AS LSA 0
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
Number of areas transit capable is 0
External flood list length 0
IETF NSF helper support enabled
Cisco NSF helper support enabled
Area BACKBONE(0)
  Number of interfaces in this area is 8 (1 loopback)
  Area has no authentication
  SPF algorithm last executed 00:39:03.224 ago
  SPF algorithm executed 3 times
  Area ranges are
  Number of LSA 15. Checksum Sum 0x087788
  Number of opaque link LSA 0. Checksum Sum 0x000000
  Number of DCbitless LSA 0
  Number of indication LSA 0
  Number of DoNotAge LSA 0
  Flood list length 0
```

IP Route OSPF:

```
R1#show ip route ospf
 2.0.0.0/32 is subnetted, 1 subnets
O       2.2.2.2 [110/2] via 10.0.0.2, 00:39:48, FastEthernet0/0
 6.0.0.0/32 is subnetted, 1 subnets
O       6.6.6.6 [110/12] via 10.0.0.2, 00:39:48, FastEthernet0/0
 7.0.0.0/32 is subnetted, 1 subnets
O       7.7.7.7 [110/11] via 10.0.0.22, 00:39:48, Ethernet2/0
 8.0.0.0/32 is subnetted, 1 subnets
O       8.8.8.8 [110/12] via 10.0.0.2, 00:39:48, FastEthernet0/0
O      192.168.20.0/24 [110/21] via 10.0.0.2, 00:39:48, FastEthernet0/0
10.0.0.0/30 is subnetted, 7 subnets
O      10.0.0.24 [110/11] via 10.0.0.2, 00:39:48, FastEthernet0/0
O      10.0.0.16 [110/11] via 10.0.0.2, 00:39:48, FastEthernet0/0
R1#
```

IP Protocols:

```
R1#show ip protocols
Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    1.1.1.1 0.0.0.0 area 0
    10.0.0.0 0.0.0.255 area 0
    172.16.100.0 0.0.0.255 area 0
    192.168.10.0 0.0.0.255 area 0
    192.168.50.0 0.0.0.255 area 0
  Reference bandwidth unit is 100 mbps
  Routing Information Sources:
    Gateway         Distance      Last Update
    2.2.2.2          110          00:40:40
    5.5.5.5          110          00:40:40
    6.6.6.6          110          00:40:40
    7.7.7.7          110          00:40:40
    8.8.8.8          110          00:40:40
  Distance: (default is 110)

Routing Protocol is "bgp 65001"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  IGP synchronization is disabled
  Automatic route summarization is disabled
  Neighbor(s):
    Address          FiltrIn FiltrOut DistIn DistOut Weight RouteMap
    10.0.0.14        CUSTOMER-A
    10.0.0.22
  Maximum path: 1
  Routing Information Sources:
    Gateway         Distance      Last Update
    10.0.0.14        20           00:40:32
  Distance: external 20 internal 200 local 200
```

4.5 BGP — Border Gateway Protocol:

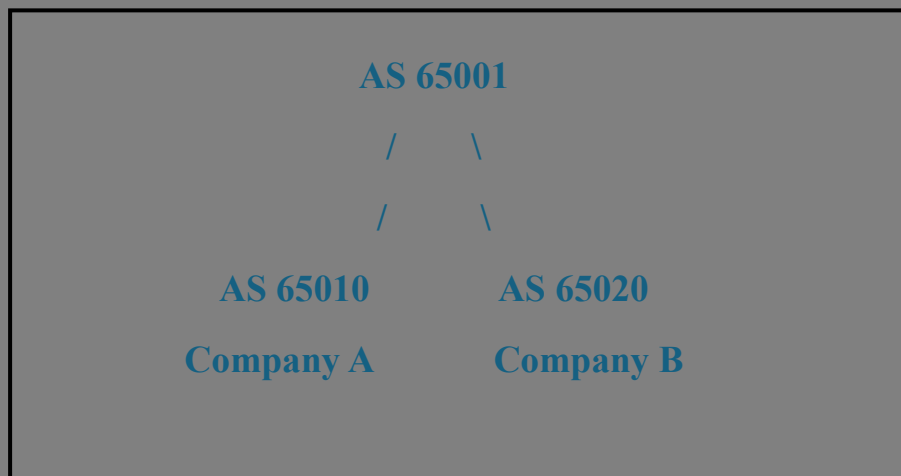
BGP is used in the project to represent routing between different Autonomous Systems.

Unlike OSPF, which is primarily used for internal routing, BGP is designed to exchange routing information between different autonomous systems and provides extensive policy-control capabilities.

The project contains multiple AS numbers representing different organizations or routing domains.

Autonomous Systems Used:

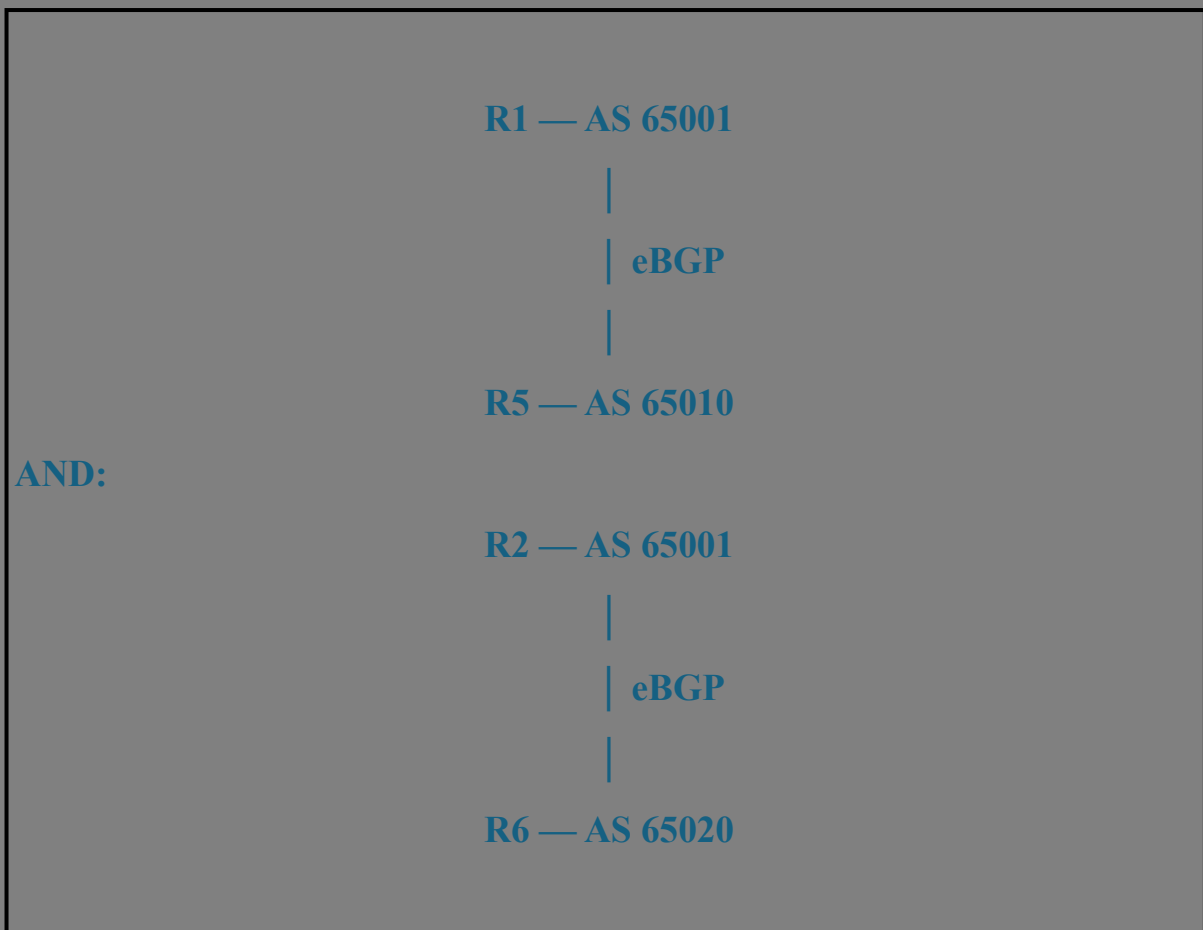
AS	Role
65001	Main ISP/Core
65010	Company A
65020	Company B



4.6 eBGP:

The BGP sessions between the different AS numbers are eBGP (External BGP) sessions.

For example:



eBGP allows the routing domains to exchange their reachable networks.

```
R1#show ip bgp summary
BGP router identifier 1.1.1.1, local AS number 65001
BGP table version is 5, main routing table version 5
4 network entries using 528 bytes of memory
5 path entries using 260 bytes of memory
4/2 BGP path/bestpath attribute entries using 672 bytes of memory
1 BGP AS-PATH entries using 24 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 1 (at peak 1) using 32 bytes of memory
BGP using 1516 total bytes of memory
BGP activity 4/0 prefixes, 5/0 paths, scan interval 60 secs

Neighbor      V      AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/
PfxRcd
10.0.0.14     4      65010    49      50       5    0    0 00:47:02
2
10.0.0.22     4      65010    49      50       5    0    0 00:47:05
1
```

4.7 BGP Prefix Lists:

Prefix lists are used to control which routes are accepted from BGP neighbors.

For example, the project uses separate prefix lists for Company A and Company B.

A simplified policy is:

```
R1#show ip prefix-list
ip prefix-list COMAPNY-A: 0 entries
ip prefix-list COMPANY-A: 2 entries
  seq 10 permit 192.168.10.0/24
  seq 20 permit 5.5.5.5/32
R1#
```

This provides route filtering and prevents unwanted prefixes from being accepted.

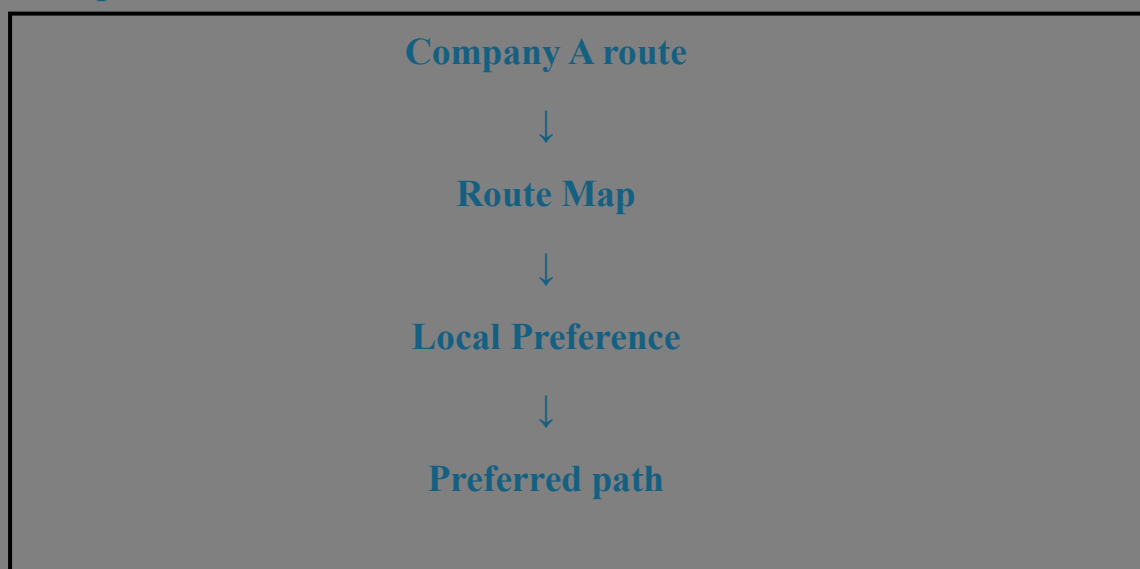
Prefix lists improve security and routing control by explicitly defining the networks that are allowed.

4.8 BGP Route Maps:

Route maps are used to apply routing policies to selected routes.

In this project, route maps are used with BGP to apply policy such as local preference.

For example:



```

R1#show route-map
route-map CUSTOMER-A, permit, sequence 10
  Match clauses:
  Set clauses:
    local-preference 200
  Policy routing matches: 0 packets, 0 bytes
R1#

```

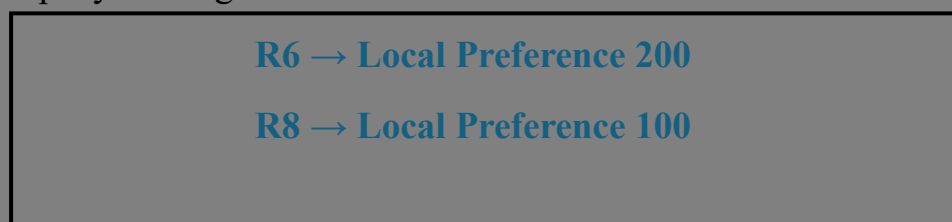
This allows the administrator to influence which available BGP path is preferred.

4.9 Local Preference and BGP Path Selection:

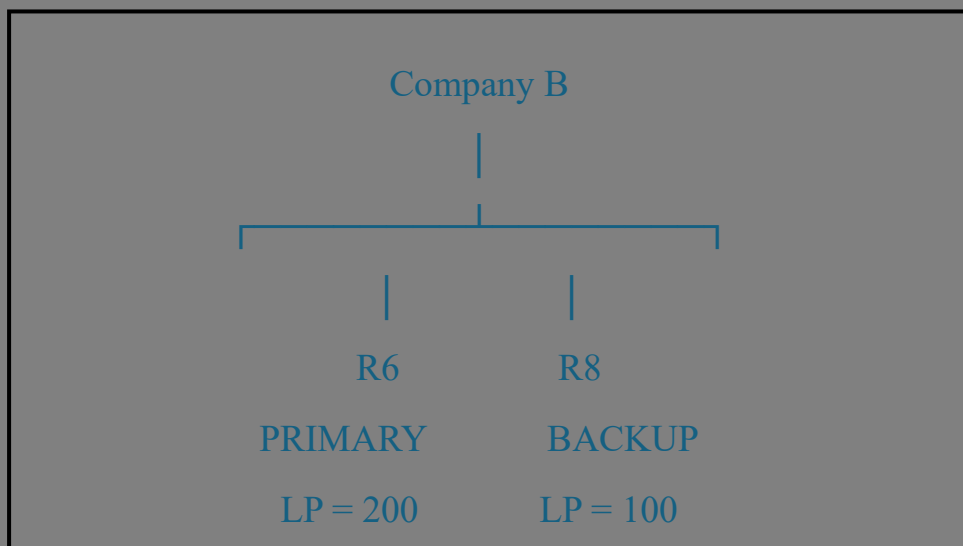
Local Preference is a BGP attribute used to influence the preferred exit path within an Autonomous System.

A higher local preference is preferred.

In the Company B design:



Therefore, R6 is preferred as the primary path while R8 provides the backup path.



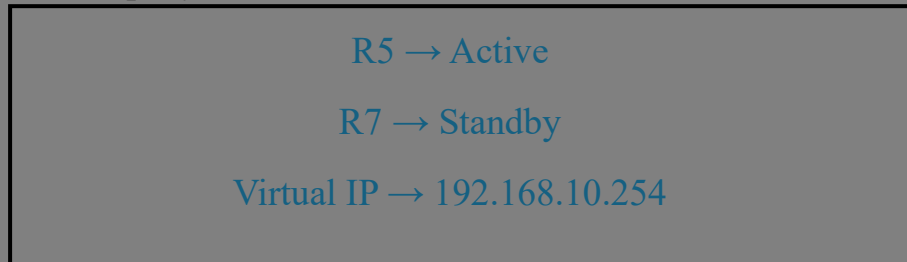
This demonstrates how BGP policy can be used to control path selection instead of relying only on physical path length.

4.10 HSRP — First Hop Redundancy:

HSRP is used to provide first-hop gateway redundancy.

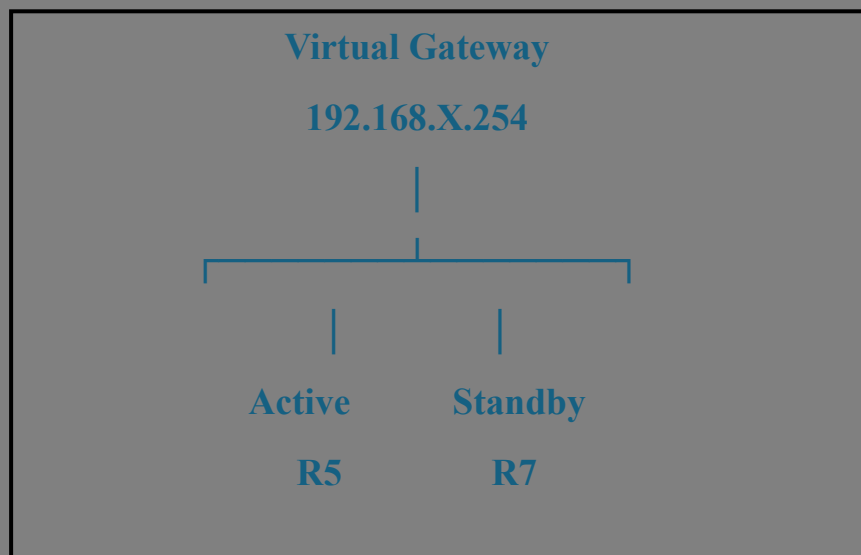
Instead of clients depending on a single physical router, HSRP provides a virtual IP address shared by an active and standby router.

For example, Company A uses:



Company B uses a similar primary/backup gateway design

HSRP Operation:



If the active router fails, the standby router takes over the virtual gateway.

4.11 HSRP Failover Testing:

HSRP failover was tested by completely shutting down the active router.

For Company A:

Before failure:

R5 → ACTIVE

R7 → STANDBY

After shutting down R5:

R5 → DOWN

R7 → ACTIVE

The virtual gateway remains available to the client.

This demonstrates that users can continue to use the virtual gateway even when the primary router fails.

R5 Is on Active Mode:

```
R5#
R5#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R5(config)#int e1/1
R5(config-if)#standby 10 ip 192.168.10.254
R5(config-if)#standby 10 priority 110
R5(config-if)#standby 10 preempt
R5(config-if)#exit
R5(config)#end
*Aug 12 23:59:46.803: %HSRP-5-STATECHANGE: Ethernet1/1 Grp 10 state Standby -> Active
R5(config)#end
R5#
*Aug 12 23:59:49.755: %SYS-5-CONFIG_I: Configured from console by console
R5#wr
Building configuration...
[OK]
R5#show standby brief
      P indicates configured to preempt.
      |
Interface Grp Pri P State Active Standby Virtual IP
Et1/1      10  110 P Active local      unknown  192.168.10.254
R5#show standby brief
      P indicates configured to preempt.
      |
Interface Grp Pri P State Active Standby Virtual IP
Et1/1      10  110 P Active local      192.168.10.2  192.168.10.254
R5#
```

R7 Is on Standby Mode:

```
% Invalid input detected at ... marker.

R7#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R7(config)#int e1/1
R7(config-if)#standby 10 ip 192.168.10.254
R7(config-if)#standby 10 priority 100
R7(config-if)#standby 10 preempt
R7(config-if)#exit
R7(config)#end
*Aug 13 00:02:01.395: %HSRP-5-STATECHANGE: Ethernet1/1 Grp 10 state Speak -> Standby
R7(config)#end
R7#
*Aug 13 00:02:03.179: %SYS-5-CONFIG_I: Configured from console by console
R7#wr
Building configuration...
[OK]
R7#show standby brief
      P indicates configured to preempt.
      |
Interface Grp Pri P State Active Standby Virtual IP
Et1/1      10  100 P Standby 192.168.10.1 local      192.168.10.254
R7#
```

HSRP Failover:

Shutting down R5 to test HSRP

```
R5(config)#int e1/1
R5(config-if)#shutdown
R5(config-if)#end
*Aug 13 00:11:43.671: %HSRP-5-STATECHANGE: Ethernet1/1 Grp 10 state Active -> In
it
R5(config-if)#end
R5#
*Aug 13 00:11:45.667: %LINK-5-CHANGED: Interface Ethernet1/1, changed state to a
dministratively down
*Aug 13 00:11:46.267: %SYS-5-CONFIG_I: Configured from console by console
*Aug 13 00:11:46.667: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/
1, changed state to down
R5#
```

Standby router becoming active

```
R7#
*Aug 13 00:11:52.799: %HSRP-5-STATECHANGE: Ethernet1/1 Grp 10 state Standby -> Active
R7#show standby brief
                P indicates configured to preempt.
                |
Interface  Grp  Pri P State  Active      Standby      Virtual IP
Et1/1      10   100 P Active local      unknown      192.168.10.254
R7#
```

4.12 BGP Failover Testing:

BGP redundancy was also tested by shutting down the primary company router.

Company A:

Primary:

R5

Backup:

R7

When R5 was stopped, the routing infrastructure learned the alternate path through R7.

Similarly,

for Company B:

Primary:

R6

Backup:

R8

When R6 was stopped, R8 provided the alternate path. This demonstrates that both the routing layer and gateway layer contain redundancy.

Primary route before failure.

```
[OK]
R1#show ip bgp summary
BGP router identifier 1.1.1.1, local AS number 65001
BGP table version is 5, main routing table version 5
4 network entries using 528 bytes of memory
5 path entries using 260 bytes of memory
4/2 BGP path/bestpath attribute entries using 672 bytes of memory
1 BGP AS-PATH entries using 24 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 1 (at peak 1) using 32 bytes of memory
BGP using 1516 total bytes of memory
BGP activity 4/0 prefixes, 5/0 paths, scan interval 60 secs

Neighbor      V      AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/
PfxRcd
10.0.0.14      4      65010     20     20       5    0    0 00:15:05
2
10.0.0.22      4      65010      4      5       5    0    0 00:00:29
1

R1#show ip bgp 192.168.10.0
BGP routing table entry for 192.168.10.0/24, version 4
Paths: (2 available, best #2, table Default-IP-Routing-Table)
  Advertised to update-groups:
    1
  65010
    10.0.0.22 from 10.0.0.22 (7.7.7.7)
      Origin IGP, metric 0, localpref 100, valid, external
  65010
    10.0.0.14 from 10.0.0.14 (5.5.5.5)
      Origin IGP, metric 0, localpref 200, valid, external, best

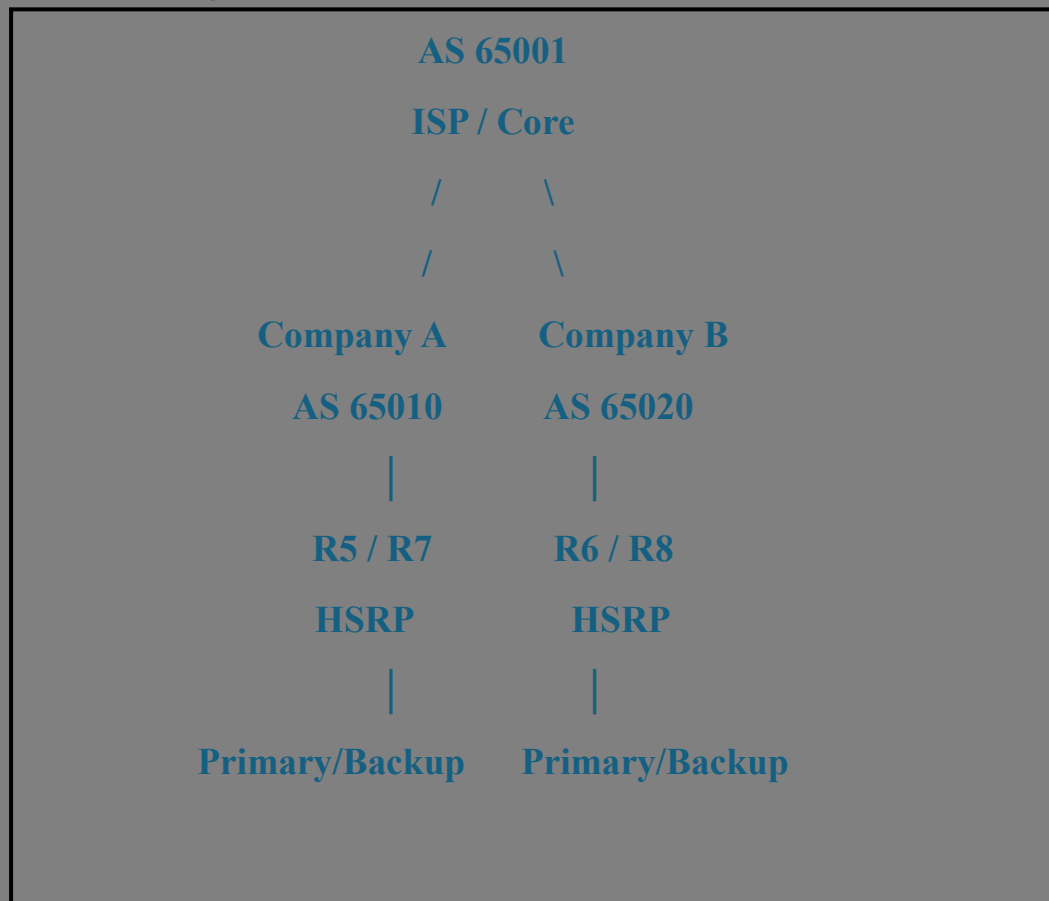
R1#show ip bgp 192.168.10.0
BGP routing table entry for 192.168.10.0/24, version 4
Paths: (2 available, best #2, table Default-IP-Routing-Table)
  Advertised to update-groups:
    1
  65010
    10.0.0.22 from 10.0.0.22 (7.7.7.7)
      Origin IGP, metric 0, localpref 100, valid, external
  65010
    10.0.0.14 from 10.0.0.14 (5.5.5.5)
      Origin IGP, metric 0, localpref 200, valid, external, best
R1#
```

Primary router is stopped, showing the backup path.

```
R1#show ip bgp 192.168.10.0
BGP routing table entry for 192.168.10.0/24, version 7
Paths: (1 available, best #1, table Default-IP-Routing-Table)
Flag: 0x840
    Not advertised to any peer
  65010
    10.0.0.22 from 10.0.0.22 (7.7.7.7)
      Origin IGP, metric 0, localpref 100, valid, external, best
R1#
```

4.13 Routing and High Availability Design:

The combined routing architecture can be summarized as:



OSPF provides internal route exchange, BGP provides inter-AS routing and routing policy, while HSRP provides first-hop gateway redundancy.

Together, these technologies improve availability, scalability, and resilience.

CHAPTER 5 — pfSense FIREWALL, NAT AND SECURITY POLICIES

5.1 Introduction

pfSense is the central security component of the proposed network. It acts as the boundary between the external network, internal enterprise networks, and the DMZ.

The firewall is responsible for controlling traffic based on source, destination, protocol, and port. In addition to firewall filtering, pfSense provides Network Address Translation (NAT), Port Address Translation (PAT), port forwarding, DNS forwarding, VPN access, and firewall logging.

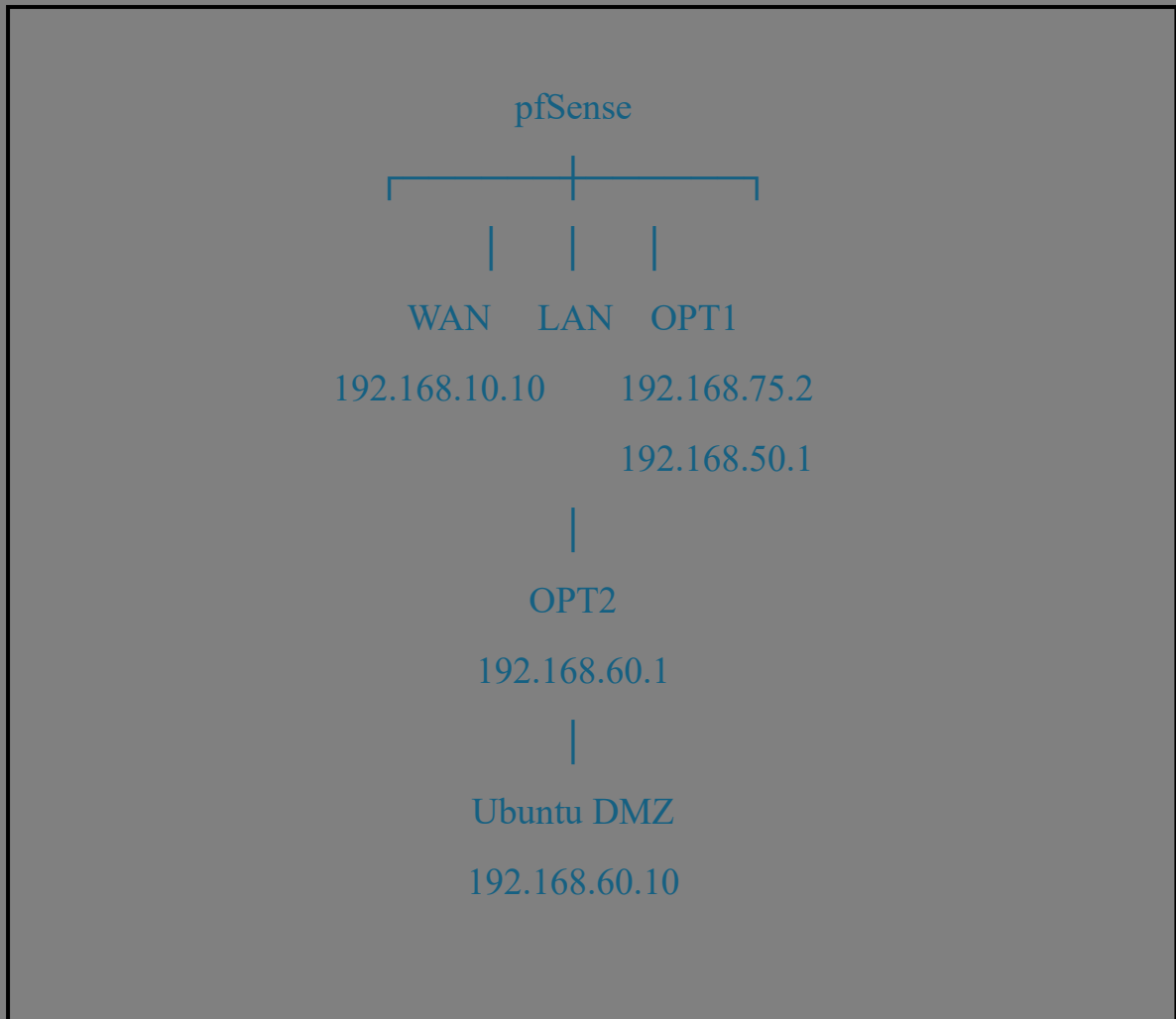
The objective of this implementation is to ensure that only required traffic is permitted while unnecessary or unauthorized communication is blocked.

5.2 pfSense Interface Architecture:

The pfSense firewall uses multiple interfaces to separate different network zones.

Interfaces   			
 WAN		1000baseT <full-duplex>	192.168.10.10
 LAN		1000baseT <full-duplex>	192.168.75.2
 OPT1		1000baseT <full-duplex>	192.168.50.1
 OPT2		1000baseT <full-duplex>	192.168.60.1

The DMZ is implemented through OPT2.



5.3 Firewall Security Zones:

The firewall separates the environment into different trust zones.

WAN

The WAN represents external traffic entering the firewall.

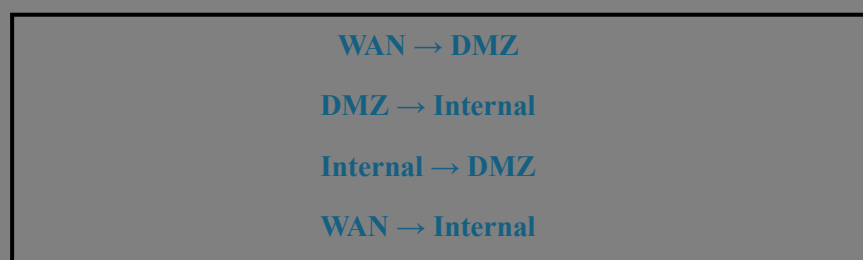
Internal Network

The internal networks contain enterprise infrastructure and company resources.

DMZ

The DMZ hosts services that require controlled access from external systems.

The firewall therefore controls communication such as:



5.4 Firewall Rule Design:

Firewall rules are configured using the principle of least privilege.

Only traffic required by the project is permitted.

WAN firewall rule:

Floating

WAN

LAN

OPT1

OPT2

OpenVPN

Rules (Drag to Change Order)

☐	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐	✓	0/224 B	IPv4 TCP	192.168.10.13	*	192.168.60.10	22 (SSH)	*	none	Allow Kali SSH to DMZ	
☐	✓	0/56 B	IPv4 UDP	192.168.10.13	*	192.168.60.10	514 (Syslog)	*	none	Allow Kali Syslog to DMZ	
☐	✓	0/224 B	IPv4 TCP	192.168.10.13	*	192.168.60.10	21 (FTP)	*	none	Allow Kali FTP to DMZ	
☐	✓	0/128 B	IPv4 TCP/UDP	192.168.10.13	*	WAN address	53 (DNS)	*	none		
☐	✓	0/0 B	IPv4 UDP	*	*	WAN address	*	*	none	Allow OpenVPN	
☐	✓	0/2 KiB	IPv4 TCP	*	*	192.168.60.10	80 (HTTP)	*	none	NAT WAN TO DMZ WEB SERVER	
☐	✓	0/6 KiB	IPv4 TCP	*	*	192.168.60.10	443 (HTTPS)	*	none	NAT WAN to DMZ HTTPS	

Add Add Delete Toggle Copy Save Separator

OPT2 rules showing DMZ-to-internal network blocking and permitted DMZ traffic.

Floating

WAN

LAN

OPT1

OPT2

OpenVPN

Rules (Drag to Change Order)

☐	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐	✗	0/336 B	IPv4 *	OPT2 subnets	*	192.168.50.0/24	*	*	none	Block DMZ to Internal Network	
☐	✗	0/0 B	IPv4 *	OPT2 subnets	*	192.168.75.0/24	*	*	none	Block DMZ To LAN	
☐	✓	0/0 B	IPv4 TCP/UDP	OPT2 subnets	*	*	*	*	none	DMZ To Internet	
☐	✓	0/0 B	IPv4 *	*	*	*	*	*	none		

Add Add Delete Toggle Copy Save Separator

5.5 Outbound NAT and PAT:

Network Address Translation is used to translate private IP addresses into an address that can communicate with external networks.

For outbound Company B traffic, the internal address:

192.168.20.100

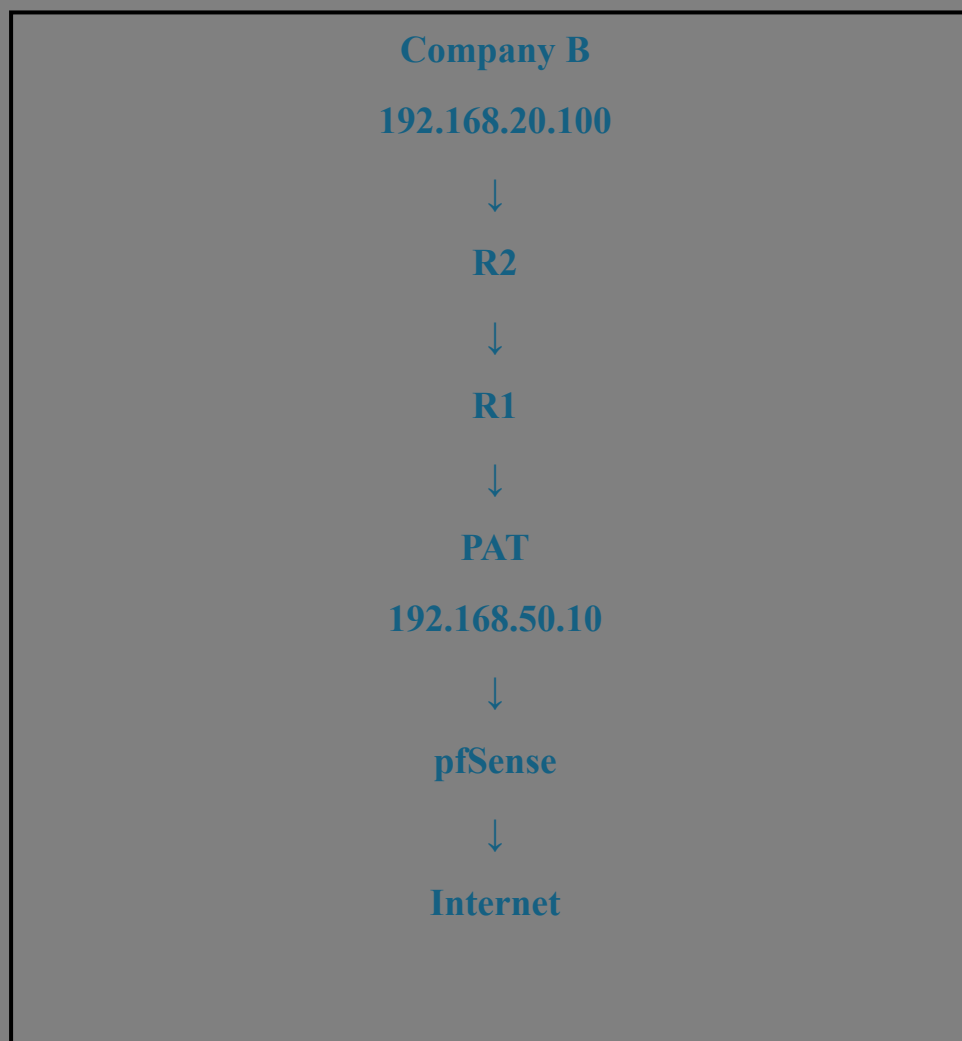
is translated by R1 to:

192.168.50.10

using PAT.

The PAT configuration allows multiple internal connections to use the same translated address by distinguishing individual sessions through port information.

The general path is:



NAT/PAT Translation Table:

Generating Traffic

```
PC2> ping 9.9.9.9
84 bytes from 9.9.9.9 icmp_seq=1 ttl=55 time=200.023 ms
84 bytes from 9.9.9.9 icmp_seq=2 ttl=55 time=152.523 ms
84 bytes from 9.9.9.9 icmp_seq=3 ttl=55 time=154.114 ms
84 bytes from 9.9.9.9 icmp_seq=4 ttl=55 time=152.693 ms
84 bytes from 9.9.9.9 icmp_seq=5 ttl=55 time=154.225 ms
```

NAT/PAT:

```
icmp 192.168.50.10:21104 192.168.20.100:21104 9.9.9.9:21104 9.9.9.9:21104
R1#show ip nat translation
Pro Inside global      Inside local      Outside local      Outside global
icmp 192.168.50.10:29552 192.168.10.100:29552 9.9.9.9:29552 9.9.9.9:29552
icmp 192.168.50.10:29808 192.168.10.100:29808 9.9.9.9:29808 9.9.9.9:29808
icmp 192.168.50.10:30064 192.168.10.100:30064 9.9.9.9:30064 9.9.9.9:30064
icmp 192.168.50.10:30320 192.168.10.100:30320 9.9.9.9:30320 9.9.9.9:30320
icmp 192.168.50.10:30576 192.168.10.100:30576 9.9.9.9:30576 9.9.9.9:30576
icmp 192.168.50.10:20080 192.168.20.100:20080 9.9.9.9:20080 9.9.9.9:20080
icmp 192.168.50.10:20336 192.168.20.100:20336 9.9.9.9:20336 9.9.9.9:20336
icmp 192.168.50.10:20592 192.168.20.100:20592 9.9.9.9:20592 9.9.9.9:20592
icmp 192.168.50.10:20848 192.168.20.100:20848 9.9.9.9:20848 9.9.9.9:20848
icmp 192.168.50.10:21104 192.168.20.100:21104 9.9.9.9:21104 9.9.9.9:21104
R1#
```

Nat Statistics:

```
R1#show ip nat statistics
Total active translations: 0 (0 static, 0 dynamic; 0 extended)
Peak translations: 10, occurred 00:01:01 ago
Outside interfaces:
  Ethernet1/3
Inside interfaces:
  FastEthernet0/0, Ethernet1/2
Hits: 20 Misses: 0
CEF Translated packets: 20, CEF Punted packets: 0
Expired translations: 10
Dynamic mappings:
-- Inside Source
[Id: 1] access-list 100 interface Ethernet1/3 refcount 0
Appl doors: 0
Normal doors: 0
Queued Packets: 0
R1#
```

5.6 Default Route to pfSense

R1 uses pfSense as the Internet-facing default gateway.

The default route is:

0.0.0.0/0 → 192.168.50.1

This allows unknown external destinations to be forwarded toward pfSense.

This was verified by successfully reaching external addresses from R1 and from the Company networks after NAT was correctly configured.

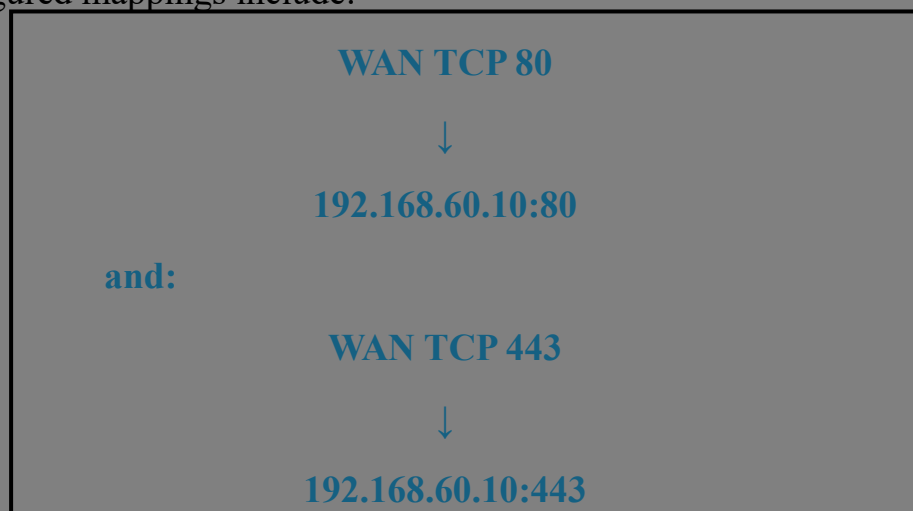
R1 Default Route Toward pfSense:

```
R1#show ip route 0.0.0.0
Routing entry for 0.0.0.0/0, supernet
  Known via "static", distance 1, metric 0, candidate default path
  Advertised by bgp 65001
  Routing Descriptor Blocks:
  * 192.168.50.1
    Route metric is 0, traffic share count is 1
```

5.7 Port Forwarding

Port forwarding is used to provide controlled access to DMZ services.

The configured mappings include:



This allows external clients to access the web server without exposing the internal network.

pfSense Port Forwarding Configuration:

Port Forward 1:1 Outbound NAT

Rules										
	Interface	Protocol	Source Address	Source Ports	Dest. Address	Dest. Ports	NAT IP	NAT Ports	Description	Actions
☐	☒ WAN	TCP	*	*	WAN address	443 (HTTPS)	192.168.60.10	443 (HTTPS)	WAN to DMZ HTTPS	  
☐	☒ WAN	TCP	*	*	WAN address	80 (HTTP)	192.168.60.10	80 (HTTP)	WAN TO DMZ WEB SERVER	  

 Add  Add  Delete  Toggle  Save  Separator

5.8 Outbound NAT

Outbound NAT controls how traffic leaving internal networks is translated when it passes through pfSense toward an external network.

In this project, pfSense was changed from Automatic Outbound NAT to Hybrid Outbound NAT so that automatically generated rules could remain available while allowing manual NAT rules to be added.

The configuration includes a rule for the enterprise network:

Interface: WAN

Source: 192.168.50.0/24

Destination: Any

Translation: WAN address

This translates traffic from the internal 192.168.50.0/24 network to the pfSense WAN address when traffic exits through the WAN interface.

Outbound NAT Mode

Mode

☐ Automatic outbound NAT rule generation. (IPsec passthrough included)

☒ Hybrid Outbound NAT rule generation. (Automatic Outbound NAT + rules below)

☐ Manual Outbound NAT rule generation. (AON - Advanced Outbound NAT)

☐ Disable Outbound NAT rule generation. (No Outbound NAT rules)

 Save

Mappings

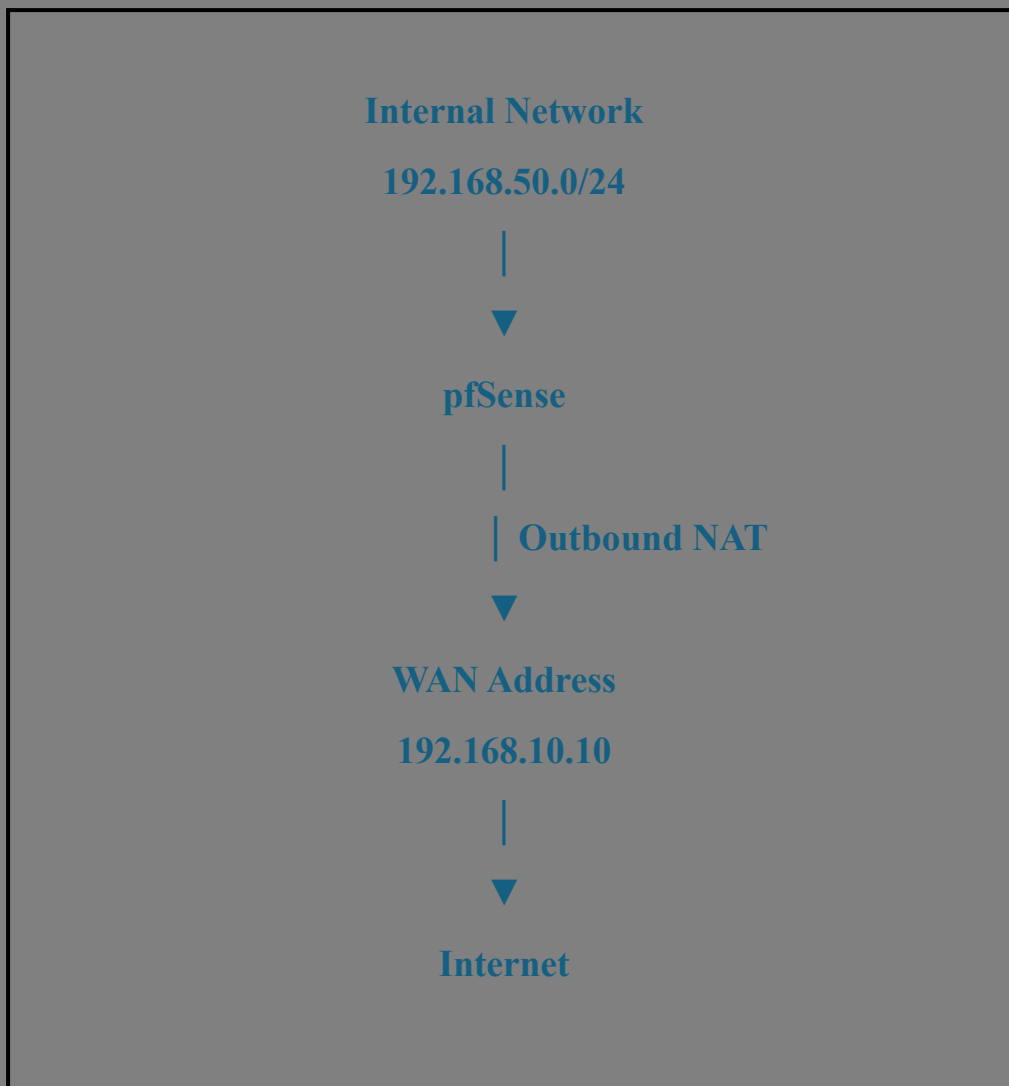
	Interface	Source	Source Port	Destination	Destination Port	NAT Address	NAT Port	Static Port	Description	Actions
☐	☒ WAN	192.168.50.0/24	*	*	*	WAN address	*		NAT-PAT Enterprise LAN	  

 Add  Add  Delete  Toggle  Save

Automatic Rules

	Interface	Source	Source Port	Destination	Destination Port	NAT Address	NAT Port	Static Port	Description
☒	WAN	127.0.0.0/8 ::1/128 192.168.10.0/24 192.168.20.0/24 10.0.0.16/30 10.0.0.12/30 192.168.50.0/24 10.10.10.0/24	*	*	500	WAN address	*	☒	Auto created rule for ISAKMP
☒	WAN	127.0.0.0/8 ::1/128 192.168.10.0/24 192.168.20.0/24 10.0.0.16/30 10.0.0.12/30 192.168.50.0/24 10.10.10.0/24	*	*	*	WAN address	*		Auto created rule
☒	LAN	127.0.0.0/8 ::1/128 192.168.10.0/24 192.168.20.0/24 10.0.0.16/30 10.0.0.12/30 192.168.50.0/24 10.10.10.0/24	*	*	500	LAN address	*	☒	Auto created rule for ISAKMP
☒	LAN	127.0.0.0/8 ::1/128 192.168.10.0/24 192.168.20.0/24 10.0.0.16/30 10.0.0.12/30 192.168.50.0/24 10.10.10.0/24	*	*	*	LAN address	*		Auto created rule

How It works:

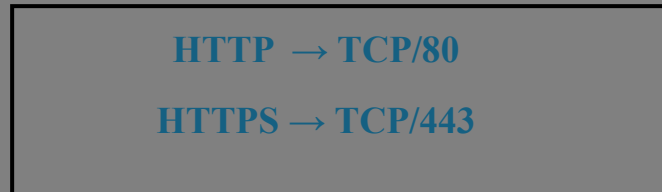


The internal private address is translated to the pfSense WAN address so that return traffic can be correctly routed back through the firewall.

5.8 DMZ Web Service Security:

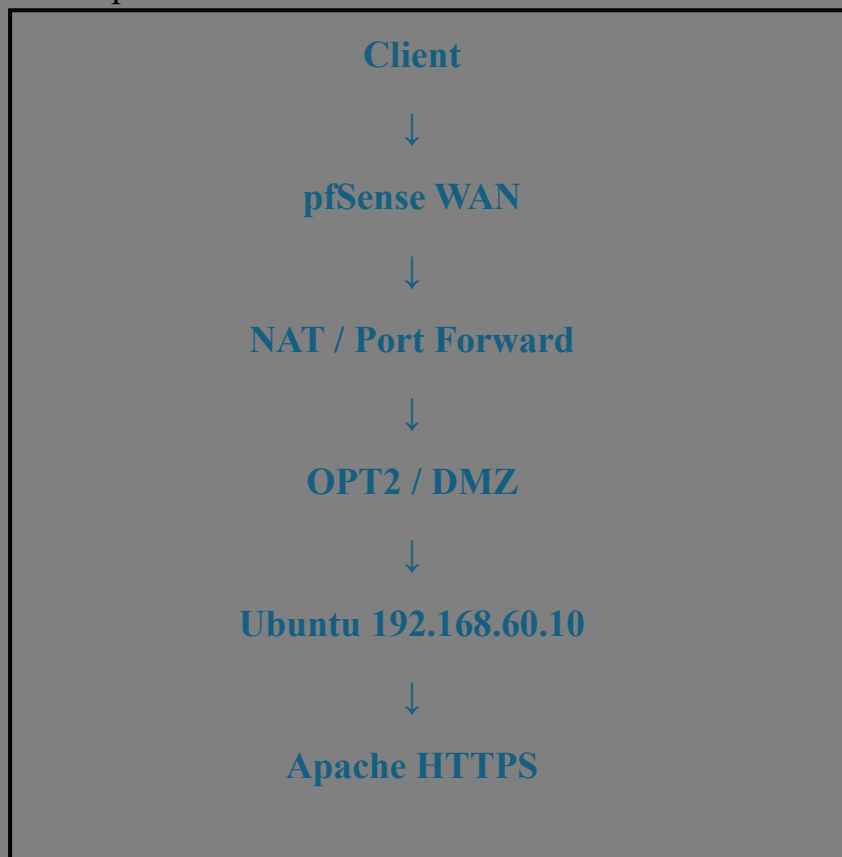
The Ubuntu web server is hosted in the DMZ rather than directly inside the internal company network.

The web service is available through:



The HTTPS service uses a self-signed SSL/TLS certificate for the laboratory environment.

The external access path is:



5.9 DNS Forwarding Through pfSense:

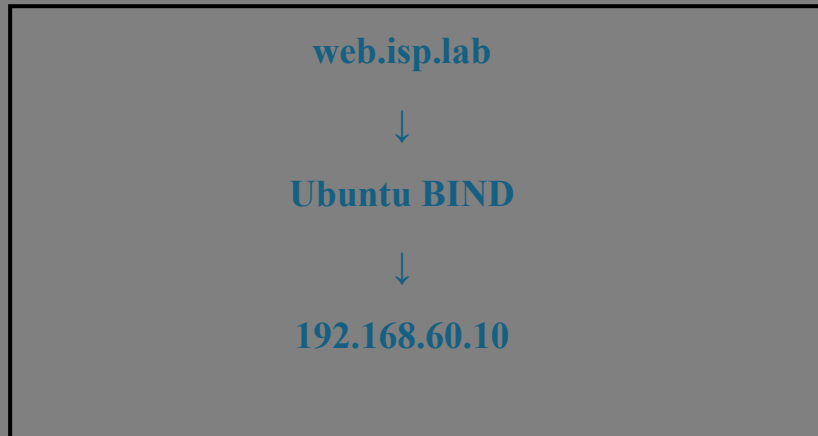
pfSense is also used to forward requests for the internal project domain.

The configured domain is: **isp.lab**

The Domain Override forwards requests to the Ubuntu BIND server:

isp.lab → 192.168.50.21

For example:



This allows clients to access the DMZ web server using a hostname instead of a raw IP address.

+ Add

Domain Overrides			
Domain	Lookup Server IP Address	Description	Actions
isp.lab	192.168.50.21	Internal ISP DNS	

Enter any domains for which the resolver's standard DNS lookup process should be overridden and a different (non-standard) lookup server should be queried instead. Non-standard, 'invalid' and local domains, and subdomains, can also be entered, such as 'test', 'nas.home.arpa', 'mycompany.localdomain', '1.168.192.in-addr.arpa', or 'somesite.com'. The IP address is treated as the authoritative lookup server for the domain (including all of its subdomains), and other lookup servers will not be queried. If there are multiple authoritative DNS servers available for a domain then make a separate entry for each, using the same domain name.

+ Add

5.10 DNS Resolver Access Control

To securely provide DNS resolution to the Kali testing system, an access-control list was configured in pfSense DNS Resolver.

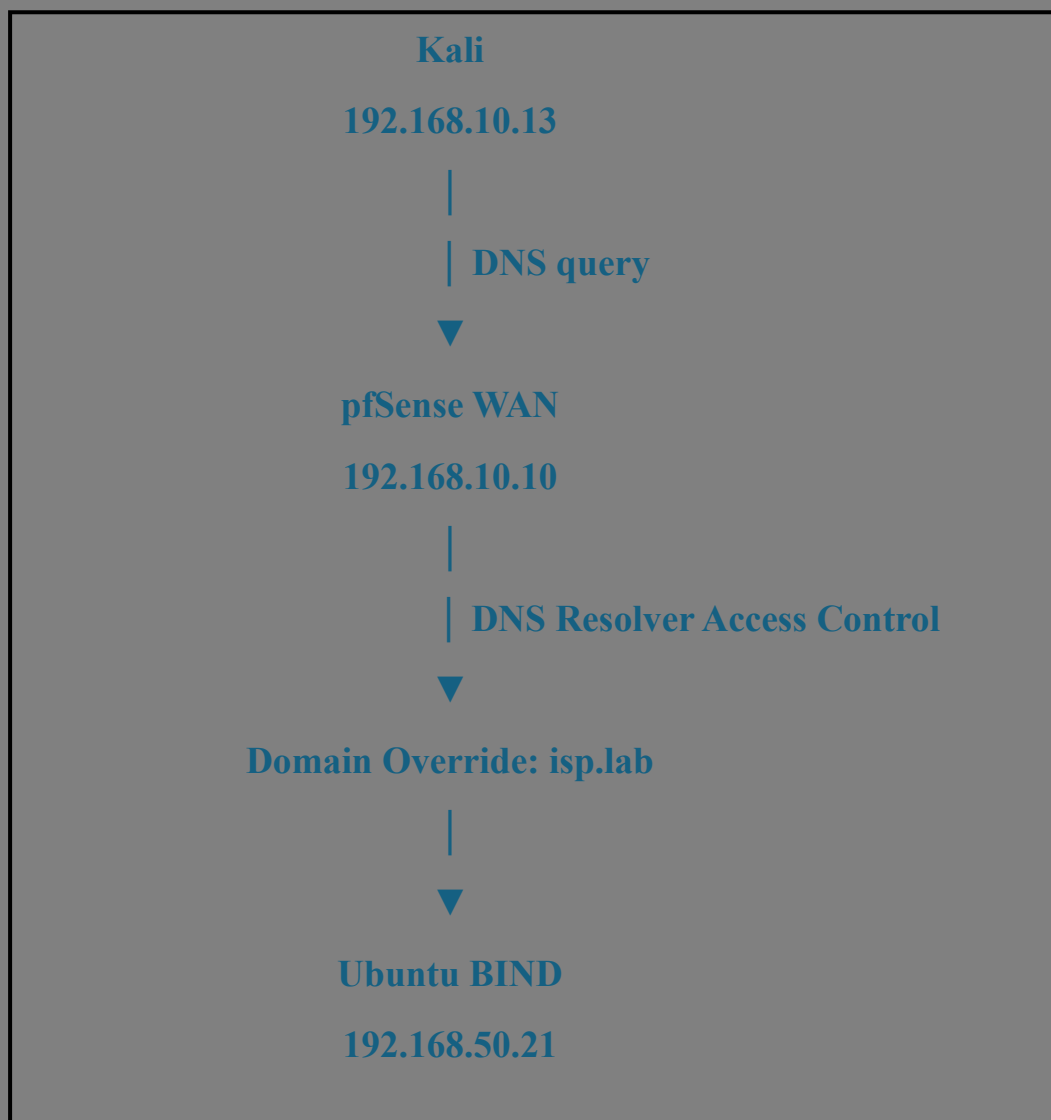
The configured access list is:

Access List Name: Allow Kali DNS

Action: Allow

The rule allows the authorized Kali client (**192.168.10.13**) to query the pfSense DNS Resolver while preventing unrestricted DNS access from other WAN clients.

The DNS flow is:



This provides controlled DNS access instead of exposing the DNS resolver to unrestricted external clients.

[General Settings](#) [Advanced Settings](#) [Access Lists](#)

Access Lists to Control Access to the DNS Resolver

Access List Name	Action	Description	Actions
Allow Kali DNS	allow		 

 Add

pfSense DNS Resolver Lookup:

Diagnostics / DNS Lookup 

DNS Lookup

Hostname

 Lookup  Add Alias

Results

Result	Record type
192.168.60.10	A

Timings

Name server	Query time
127.0.0.1	1 msec
::1	0 msec
192.168.10.1	139 msec
8.8.8.8	74 msec
1.1.1.1	64 msec

More Information

[Ping](#)

[Traceroute](#)

5.11 Firewall Logging:

pfSense firewall logging is used to record network traffic handled by firewall rules.

The logs can be used to identify:

- Allowed connections
- Blocked connections
- Source addresses
- Destination addresses
- Protocols and ports
- Security-policy events

This provides useful visibility during troubleshooting and security testing.

✗	Aug 15 23:00:36	WAN	Default deny rule IPv4 (1000000103)	i 192.168.10.9:138	i 192.168.10.255:138	UDP
✗	Aug 15 23:00:36	OPT2	Default deny rule IPv4 (1000000103)	i 192.168.239.1:138	i 192.168.239.255:138	UDP
✗	Aug 15 23:01:07	WAN	Default deny rule IPv4 (1000000103)	i 192.168.10.13:47616	i 192.168.60.10:3306	TCP:S
✗	Aug 15 23:01:08	WAN	Default deny rule IPv4 (1000000103)	i 192.168.10.13:47616	i 192.168.60.10:3306	TCP:S
✗	Aug 15 23:01:09	WAN	Default deny rule IPv4 (1000000103)	i 192.168.10.13:47616	i 192.168.60.10:3306	TCP:S
✗	Aug 15 23:01:10	WAN	Default deny rule IPv4 (1000000103)	i 192.168.10.13:47616	i 192.168.60.10:3306	TCP:S
✗	Aug 15 23:01:11	WAN	Default deny rule IPv4 (1000000103)	i 192.168.10.13:47616	i 192.168.60.10:3306	TCP:S

5.11 DMZ Security Verification:

The firewall configuration was tested to ensure that the DMZ can provide required services while being restricted from protected networks.

Testing demonstrated:

DMZ → Internet	✓
DMZ → Internal network	✗
WAN → HTTP	✓
WAN → HTTPS	✓
Kali → FTP	✓
Kali → SSH	✓
Kali → Syslog	✓
Unwanted services	✗

The DMZ isolation was specifically tested by forcing traffic from the DMZ interface toward the internal network and confirming that the firewall blocked the connection.

DMZ To Internal Network:

```
haider@linux-server:~$ ping -c 4 192.168.50.10
PING 192.168.50.10 (192.168.50.10) 56(84) bytes of data.

--- 192.168.50.10 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3107ms

haider@linux-server:~$ _
```

5.12 WAN Exposure Verification

Nmap was used from Kali to identify services exposed on the pfSense WAN interface.

The verification identified the intentionally exposed services

Including:

80/tcp HTTP

443/tcp HTTPS

while unnecessary services such as FTP, SSH, and Syslog were restricted or filtered.

The DNS service on the firewall was separately reviewed because DNS access was intentionally configured for the Kali test client.

```
(kali@kali)-[~]
$ nmap -Pn -p 80,443,21,22,53,514 192.168.10.10
Starting Nmap 7.98 ( https://nmap.org ) at 2026-08-15 13:56 -0400
Nmap scan report for pfSense (192.168.10.10)
Host is up (0.00077s latency).

PORT      STATE      SERVICE
21/tcp    filtered  ftp
22/tcp    filtered  ssh
53/tcp    open      domain
80/tcp    open      http
443/tcp   open      https
514/tcp   filtered  shell
MAC Address: 00:0C:29:39:EA:17 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 1.43 seconds

(kali@kali)-[~]
$
```

5.13 Security Design Principles:

The pfSense configuration follows several security principles:

Least Privilege:

Only required services are permitted.

Network Segmentation:

Internal networks and the DMZ are separated.

Controlled Exposure:

Only HTTP and HTTPS are intentionally exposed as public-facing web services.

Restricted Administration:

SSH access is restricted to the authorized testing source in the laboratory.

Logging:

Firewall activity is recorded for analysis and troubleshooting.

CHAPTER 6 — DMZ AND SERVER SERVICES

6.1 Introduction:

The DMZ is a dedicated network segment used to host services that require controlled access from external or less-trusted networks.

In this project, pfSense provides the DMZ gateway through OPT2, while the Ubuntu Server provides the services hosted in the DMZ.

The DMZ network is:

192.168.60.0/24

with:

pfSense OPT2 → 192.168.60.1

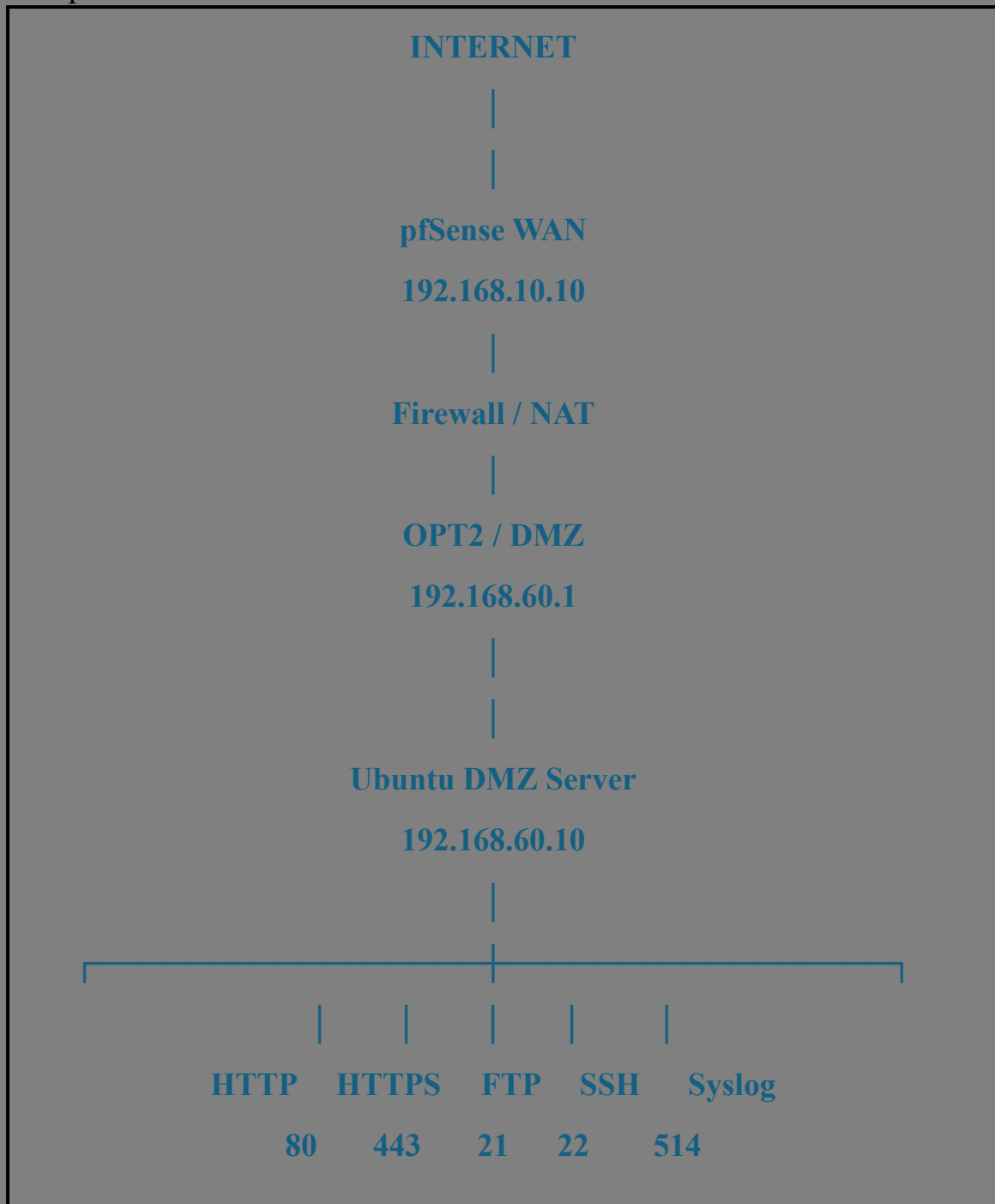
Ubuntu DMZ → 192.168.60.10

The Ubuntu Server provides multiple services required by the enterprise environment, including Apache Web Server, HTTPS, BIND9 DNS, FTP, SSH, and Syslog.

The services are controlled by pfSense firewall policies and are tested from Kali Linux to verify accessibility and security.

6.2 DMZ Architecture

The implemented DMZ architecture is:



6.4 Apache Web Server:

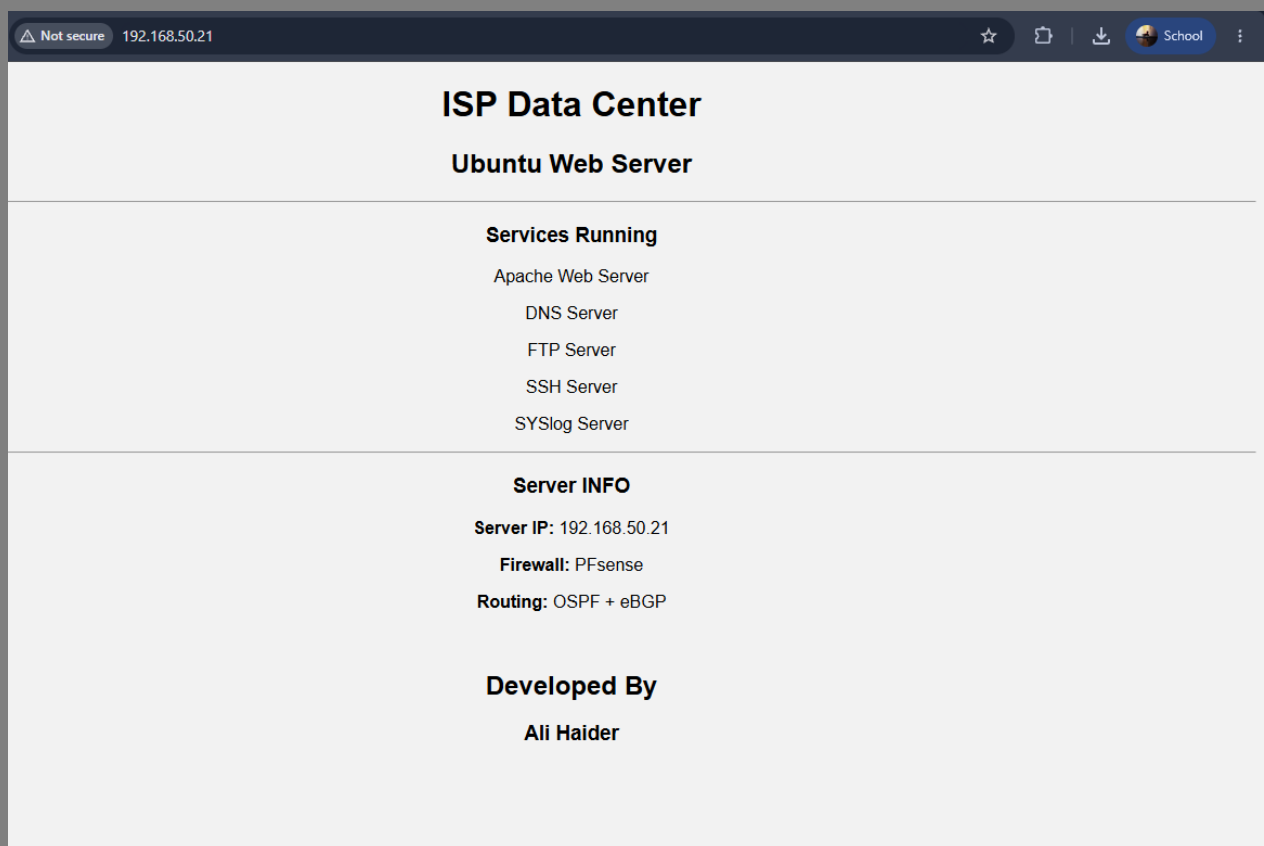
Apache2 is deployed as the primary web server in the DMZ.

The HTTP service listens on: TCP/80

The server provides a custom webpage representing the simulated ISP/Data Center environment.

The webpage includes information about the server and the services operating on the Ubuntu Server.

The service was verified locally and through the pfSense port-forward configuration.



6.5 HTTPS and SSL/TLS

HTTPS was configured to provide encrypted web communication.

Apache was configured with SSL support and a self-signed certificate for the laboratory environment.

The HTTPS service listens on: TCP/443

The certificate and Apache SSL configuration allow clients to securely access the DMZ web server.

The service was verified using:

curl -k https://192.168.60.10

The -k option was used because the certificate is self-signed and therefore not trusted by default by the client.

Apache HTTPS Configuration:

```
haider@linux-server:~$ ss -lntp | grep ':443'
LISTEN 0      511          *:*          *:*
haider@linux-server:~$
```

HTTPS Web Server Verification:

```
(kali㉿kali)-[~]
└─$ curl -k https://192.168.10.10
<!DOCTYPE HTML>
<html>
<head>
  <title> Internet Service Provider DATA CENTER</title>
</head>
<body style = "background-color:#f2f2f2; text-align:center; font-family:Arial;">
  <h1> ISP Data Center</h1>
  <h2> Ubuntu Web Server </h2>
  <hr>
  <h3> Services Running</h3>
  <p> Apache Web Server </p>
  <p> DNS Server </p>
  <p> FTP Server </p>
  <p> SSH Server </p>
  <p> SYSlog Server </p>
  <hr>
  <h3> Server INFO</h3>
  <p><b> Server IP:</b> 192.168.50.21</p>
  <p><b> Firewall:</b> PfSense </p>
  <p><b> Routing:</b> </p> OSPF + eBGP </p>
  <br>
  <h2> Developed By </h2>
  <h3> Ali Haider </h3>
</body>
</html>
```

6.6 BIND9 DNS Server

BIND9 is configured as the DNS server for the project.

A private DNS zone was created using: isp.lab

The zone contains records for the project services.

Important records include:

ns1.isp.lab → 192.168.50.21

web.isp.lab → 192.168.60.10

www.isp.lab → 192.168.60.10

ftp.isp.lab → 192.168.50.21

The DNS zone was validated using:

sudo named-checkzone isp.lab /etc/bind/db.isp.lab

The validation returned: OK

Direct DNS resolution was also verified: dig @127.0.0.1 web.isp.lab

which returned: 192.168.60.10

BIND9 DNS Resolution:

```
haider@linux-server:~$ sudo named-checkzone isp.lab /etc/bind/db.isp.local
zone isp.lab/IN: loaded serial 3
OK
haider@linux-server:~$ dig @127.0.0.1 web.isp.lab

; <<>> Dig 9.18.39-0ubuntu0.24.04.5-Ubuntu <<>> @127.0.0.1 web.isp.lab
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 34472
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: a829c630b058e6fe010000006a7f5900f0f1510eb90986b5 (good)
;; QUESTION SECTION:
;web.isp.lab.                IN      A

;; ANSWER SECTION:
web.isp.lab.                604800  IN      A      192.168.60.10

;; Query time: 0 msec
;; SERVER: 127.0.0.1#53(127.0.0.1) (UDP)
;; WHEN: Fri Aug 14 18:05:52 UTC 2026
;; MSG SIZE rcvd: 84

haider@linux-server:~$ _
```

BIND9 DNS Zone Validation:

```
haider@linux-server:~$ sudo named-checkzone isp.lab /etc/bind/db.isp.lab
[sudo] password for haider:
zone isp.lab/IN: loaded serial 3
OK
haider@linux-server:~$
```

6.8 FTP Server

The Ubuntu Server uses vsftpd to provide FTP functionality.

The FTP service listens on: TCP/21

Anonymous FTP access was disabled, and local user authentication was configured.

The important configuration includes:

anonymous_enable=NO

local_enable=YES

write_enable=YES

chroot_local_user=YES

A configuration error caused by an unsupported local_mask parameter was identified and removed. After correcting the configuration, the vsftpd service was successfully started.

FTP Service Status:

```
haider@linux-server:~$ sudo ss -ltnp | grep ":21"
LISTEN 0      32            *:21          *:21    users:(("vsftpd",pid=3210,fd=3))

haider@linux-server:~$ sudo systemctl status vsftpd
• vsftpd.service - vsftpd FTP server
   Loaded: loaded (/usr/lib/systemd/system/vsftpd.service; enabled; preset: enabled)
   Active: active (running) since Fri 2025-08-14 18:27:54 UTC; 1min 8s ago
   Process: 3208 ExecStartPre=/bin/mkdir -p /var/run/vsftpd/empty (code=exited, status=0/SUCCESS)
   Main PID: 3210 (vsftpd)
   Tasks: 1 (limit: 6537)
   Memory: 720.0K (peak: 1.5M)
   CPU: 10ms
   CGroup: /system.slice/vsftpd.service
           └─3210 /usr/sbin/vsftpd /etc/vsftpd.conf

Aug 14 18:27:54 linux-server systemd[1]: Starting vsftpd.service - vsftpd FTP server...
Aug 14 18:27:54 linux-server systemd[1]: Started vsftpd.service - vsftpd FTP server.
haider@linux-server:~$
```

FTP Port Verification:

```
(kali@kali)-[~]
$ sudo ip route add 192.168.60.0/24 via 192.168.10.10
[sudo] password for kali:

(kali@kali)-[~]
$ ip route
default via 192.168.10.1 dev eth0 proto dhcp src 192.168.10.13 metric 100
192.168.10.0/24 dev eth0 proto kernel scope link src 192.168.10.13 metric 100
192.168.60.0/24 via 192.168.10.10 dev eth0

(kali@kali)-[~]
$ nmap -Pn -p 21 192.168.60.10
Starting Nmap 7.98 ( https://nmap.org ) at 2026-08-14 14:38 -0400
Nmap scan report for 192.168.60.10
Host is up (0.0023s latency).

PORT      STATE SERVICE
21/tcp    open  ftp

Nmap done: 1 IP address (1 host up) scanned in 0.60 seconds

(kali@kali)-[~]
$
```

6.9 SSH Service

OpenSSH is enabled on the Ubuntu Server to provide secure remote administration.

The SSH service uses: TCP/22

Access from the Kali test machine was controlled through a pfSense firewall rule.

The configured rule restricts SSH access to the authorized Kali source:
192.168.10.13

This demonstrates the use of source-specific firewall rules instead of allowing unrestricted administrative access.

SSH Service Status:

```
haider@linux-server:~$ sudo systemctl status ssh
• ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Fri 2026-08-14 16:05:45 UTC; 2h 58min ago
   TriggeredBy: • ssh.socket
     Docs: man:sshd(8)
           man:sshd_config(5)
    Main PID: 1074 (sshd)
      Tasks: 1 (limit: 6537)
     Memory: 2.1M (peak: 2.4M)
        CPU: 33ms
    CGroup: /system.slice/ssh.service
            └─1074 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Aug 14 16:05:45 linux-server systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Aug 14 16:05:45 linux-server sshd[1074]: Server listening on 0.0.0.0 port 22.
Aug 14 16:05:45 linux-server sshd[1074]: Server listening on :: port 22.
Aug 14 16:05:45 linux-server systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
haider@linux-server:~$
```

SSH Service Verification:

```
haider@linux-server: ~  
Session Actions Edit View Help  
  
(kali@kali)-[~]  
$ nmap -Pn -p 22 192.168.60.10  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-08-14 15:08 -0400  
Nmap scan report for 192.168.60.10  
Host is up (0.0012s latency).  
  
PORT      STATE SERVICE  
22/tcp    open  ssh  
  
Nmap done: 1 IP address (1 host up) scanned in 0.63 seconds  
  
(kali@kali)-[~]  
$ ssh haider@192.168.60.10  
The authenticity of host '192.168.60.10 (192.168.60.10)' can't be established.  
ED25519 key fingerprint is: SHA256:CDDtcQiQWtUInVC59j55s09GA2UB1oDLz7xKiqhw3s  
This key is not known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? y  
Please type 'yes', 'no' or the fingerprint: yes  
Warning: Permanently added '192.168.60.10' (ED25519) to the list of known hosts.  
haider@192.168.60.10's password:  
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-137-generic x86_64)  
  
* Documentation:  https://help.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:       https://ubuntu.com/pro  
  
System information as of Fri Aug 14 07:08:51 PM UTC 2026  
  
System load:  0.0           Processes:      246  
Usage of /:   28.9% of 11.21GB Users logged in: 1  
Memory usage: 7%           IPv4 address for ens33: 192.168.50.21  
Swap usage:   0%  
  
* Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s  
just raised the bar for easy, resilient and secure K8s cluster deployment.  
  
https://ubuntu.com/engage/secure-kubernetes-at-the-edge  
  
Expanded Security Maintenance for Applications is not enabled.  
  
37 updates can be applied immediately.  
1 of these updates is a standard security update.  
To see these additional updates run: apt list --upgradable  
  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
  
Last login: Thu Jul 2 08:51:38 2026 from 192.168.50.11  
haider@linux-server:~$ pwd  
/home/haider  
haider@linux-server:~$ ls  
haider@linux-server:~$ ls -la  
.. .bash_history .bash_logout .bashrc .cache .lessht .profile .ssh .sudo_as_admin_successful  
haider@linux-server:~$
```

6.10 Syslog and Centralized Logging:

The Ubuntu Server runs rsyslog for system and network logging.

The service listens for remote Syslog messages on: UDP/514

The service was verified using: `sudo ss -ltnp | grep ':514'`

which showed rsyslog listening on UDP port 514.

Local logging was verified using logger TEST: Ubuntu rsyslog is working
and the message was observed in: `/var/log/syslog`

rsyslog Service Status:

```
haider@linux-server:~$ systemctl status rsyslog
● rsyslog.service - System Logging Service
   Loaded: loaded (/usr/lib/systemd/system/rsyslog.service; enabled; preset: enabled)
   Active: active (running) since Sun 2026-08-16 07:57:31 UTC; 1h 18min ago
   TriggeredBy: ● syslog.socket
     Docs: man:rsyslogd(8)
           man:rsyslog.conf(5)
           https://www.rsyslog.com/doc/
   Process: 866 ExecStartPre=/usr/lib/rsyslog/reload-apparmor-profile (code=exited, status=0/SUCCESS)
   Main PID: 890 (rsyslogd)
     Tasks: 5 (limit: 6537)
   Memory: 3.6M (peak: 5.2M)
     CPU: 247ms
   CGroup: /system.slice/rsyslog.service
           └─890 /usr/sbin/rsyslogd -n -iNONE

Aug 16 07:57:31 linux-server systemd[1]: Starting rsyslog.service - System Logging Service...
Aug 16 07:57:31 linux-server rsyslogd[890]: imuxsock: Acquired UNIX socket '/run/systemd/journal/syslog' (fd 3) from systemd. [v8.2312.0]
Aug 16 07:57:31 linux-server rsyslogd[890]: rsyslogd's groupid changed to 104
Aug 16 07:57:31 linux-server rsyslogd[890]: rsyslogd's userid changed to 103
Aug 16 07:57:31 linux-server rsyslogd[890]: [origin software="rsyslogd" swVersion="8.2312.0" x-pid="890" x-info="https://www.rsyslog.com"] start
Aug 16 07:57:31 linux-server systemd[1]: Started rsyslog.service - System Logging Service.
Aug 16 07:57:32 linux-server systemd[1]: rsyslog.service: Sent signal SIGHUP to main process 890 (rsyslogd) on client request.
Aug 16 07:57:32 linux-server rsyslogd[890]: [origin software="rsyslogd" swVersion="8.2312.0" x-pid="890" x-info="https://www.rsyslog.com"] rsyslogd was HUPed
haider@linux-server:~$
```

Remote logging was then tested from Kali.

```
haider@linux-server:~$ sudo tail -n 20 /var/log/syslog
2026-08-14T18:34:27.803059+00:00 linux-server fwupd[3239]: 18:34:27.802 FuMain fwupd 1.9.93 ready for requests (locale en_US.UTF-8)
2026-08-14T18:34:27.804795+00:00 linux-server dbus-daemon[846]: [system] Successfully activated service 'org.freedesktop.fwupd'
2026-08-14T18:34:27.804856+00:00 linux-server systemd[1]: Started fwupd.service - Firmware update daemon.
2026-08-14T18:34:27.817891+00:00 linux-server systemd[1]: fwupd-refresh.service: Deactivated successfully.
2026-08-14T18:34:27.818214+00:00 linux-server systemd[1]: Finished fwupd-refresh.service - Refresh fwupd metadata and update motd.
2026-08-14T18:35:01.289660+00:00 linux-server CRON[3250]: (root) CMD (command -v debian-sa1 > /dev/null && debian-sa1 1 1)
2026-08-14T18:39:28.354022+00:00 linux-server systemd[1]: fwupd.service: Deactivated successfully.
2026-08-14T18:40:01.226152+00:00 linux-server systemd[1]: Starting sysstat-collect.service - system activity accounting tool...
2026-08-14T18:40:01.231048+00:00 linux-server systemd[1]: sysstat-collect.service: Deactivated successfully.
2026-08-14T18:40:01.231334+00:00 linux-server systemd[1]: Finished sysstat-collect.service - system activity accounting tool.
2026-08-14T18:45:01.303141+00:00 linux-server CRON[3270]: (root) CMD (command -v debian-sa1 > /dev/null && debian-sa1 1 1)
2026-08-14T18:50:17.652760+00:00 linux-server systemd[1]: Starting sysstat-collect.service - system activity accounting tool...
2026-08-14T18:50:17.688505+00:00 linux-server systemd[1]: sysstat-collect.service: Deactivated successfully.
2026-08-14T18:50:17.688903+00:00 linux-server systemd[1]: Finished sysstat-collect.service - system activity accounting tool.
2026-08-14T18:54:39.102834+00:00 linux-server haider: TEST: Ubuntu rsyslog is working NiGGA
2026-08-14T18:55:01.323234+00:00 linux-server CRON[3319]: (root) CMD (command -v debian-sa1 > /dev/null && debian-sa1 1 1)
2026-08-14T19:00:12.805354+00:00 linux-server systemd[1]: Starting sysstat-collect.service - system activity accounting tool...
2026-08-14T19:00:12.811207+00:00 linux-server systemd[1]: sysstat-collect.service: Deactivated successfully.
2026-08-14T19:00:12.811390+00:00 linux-server systemd[1]: Finished sysstat-collect.service - system activity accounting tool.
2026-08-14T15:00:41.239670-04:00 kali kali TEST: Kali remote syslog to Ubuntu DMZ
```

6.13 DMZ Security Verification

The DMZ was tested using Kali Linux to verify both permitted and restricted services.

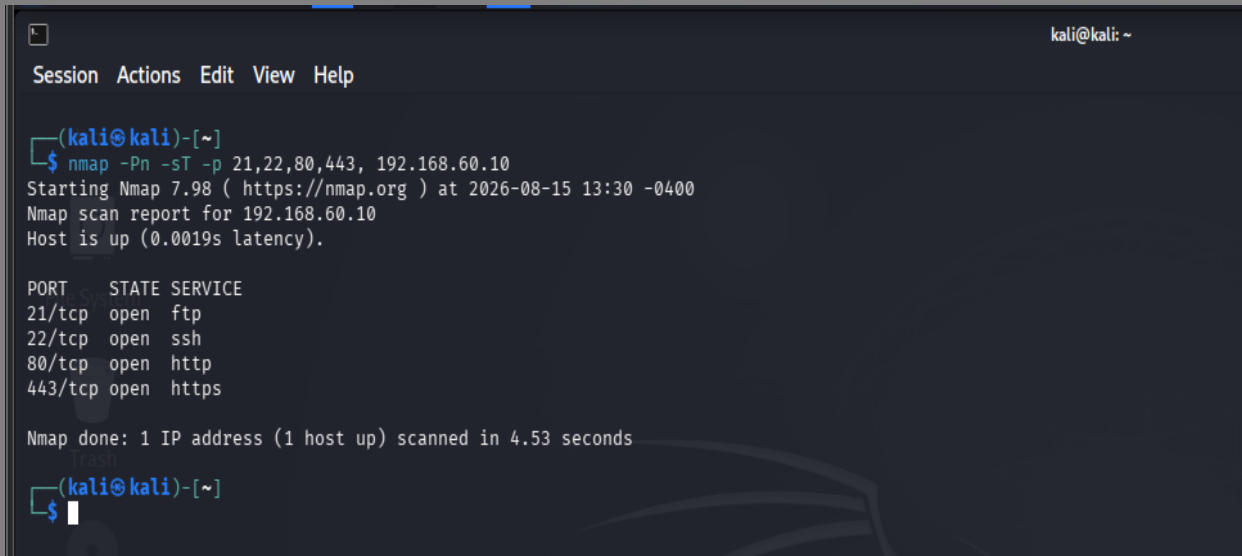
A TCP scan confirmed the intentionally accessible services:

21/tcp	FTP	open
22/tcp	SSH	open
80/tcp	HTTP	open
443/tcp	HTTPS	open

Unnecessary ports were scanned separately and were found to be closed or filtered.

The DMZ was also tested for access toward protected internal networks. By forcing traffic through the DMZ interface, communication toward the internal 192.168.50.0/24 network was successfully blocked by pfSense.

DMZ Allowed Services Scan:



```
(kali㉿kali)-[~]
$ nmap -Pn -sT -p 21,22,80,443, 192.168.60.10
Starting Nmap 7.98 ( https://nmap.org ) at 2026-08-15 13:30 -0400
Nmap scan report for 192.168.60.10
Host is up (0.0019s latency).

PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
80/tcp    open  http
443/tcp   open  https

Nmap done: 1 IP address (1 host up) scanned in 4.53 seconds
(kali㉿kali)-[~]
$
```

6.14 Architectural Limitation:

The Ubuntu Server currently uses two network interfaces:

ens33 → 192.168.50.21

ens37 → 192.168.60.10

Therefore, the server is dual-homed

Traffic generated through ens33 can directly reach the existing internal network without traversing the OPT2 firewall policy. During testing, this behavior was identified and documented.

The DMZ firewall isolation was successfully demonstrated when traffic was explicitly routed through the DMZ interface ens37.

For a production implementation, a dedicated single-homed DMZ server would provide stronger network isolation.

CHAPTER 7 — VPN AND REMOTE ACCESS SECURITY

7.1 Introduction:

Remote access is an important requirement in modern enterprise networks because authorized users may need to access internal resources from outside the organization.

In this project, OpenVPN is implemented on pfSense to provide secure remote access to the enterprise network. The VPN uses encrypted communication between the remote client and the pfSense firewall.

The OpenVPN server operates on the pfSense WAN interface and uses certificate-based security together with user authentication.

7.2 OpenVPN Server Configuration:

The pfSense OpenVPN server is configured in **Remote Access (SSL/TLS + User Authentication)** mode.

The main configuration is:

Parameter	Configuration
Interface	WAN
Protocol	UDP
Port	1194
Tunnel Network	10.10.10.0/24
Mode	Remote Access
Authentication	SSL/TLS + User Authentication
Digest	SHA256
DH Parameters	2048 bits

The VPN tunnel network provides a separate logical address space for connected VPN client

Servers	Clients	Client Specific Overrides	Wizards	Client Export
OpenVPN Servers				
Interface	Protocol / Port	Tunnel Network	Mode / Crypto	Description Actions
WAN	UDP4 / 1194 (TUN)	10.10.10.0/24	Mode: Remote Access (SSL/TLS + User Auth) Data Ciphers: AES-256-GCM, AES-128-GCM, CHACHA20-POLY1305, AES-256-CBC Digest: SHA256 D-H Params: 2048 bits	internal-ca   
				 Add

7.3 OpenVPN Authentication

The OpenVPN implementation uses **SSL/TLS certificates together with username/password authentication.**

This provides two layers of authentication:

1. Certificate-based trust between the VPN client and server.
2. User authentication for the individual VPN account.

This prevents unauthorized users from accessing the VPN simply by knowing the server address and port.

7.4 VPN Tunnel Network

A dedicated tunnel network is configured: 10.10.10.0/24

When a client connects successfully, it receives an address from the VPN tunnel network.

Remote Client



Encrypted OpenVPN Tunnel



pfSense WAN



10.10.10.0/24

- Internal Networks
- Enterprise Resources
- Authorized Services

This separates VPN client addressing from the physical LAN and DMZ networks.

7.5 OpenVPN Firewall Access:

The pfSense WAN firewall contains a rule allowing OpenVPN traffic to the configured VPN port.

The required traffic is: UDP/1194

Only this specific VPN service is exposed for remote-access establishment.

OpenVPN WAN Firewall Rule:

Rules (Drag to Change Order)											
☐	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐	✓ 0/224 B	IPv4 TCP	192.168.10.13	*	192.168.60.10	22 (SSH)	*	none		Allow Kali SSH to DMZ	Anchor Edit Copy Delete Toggle
☐	✓ 0/56 B	IPv4 UDP	192.168.10.13	*	192.168.60.10	514 (Syslog)	*	none		Allow Kali Syslog to DMZ	Anchor Edit Copy Delete Toggle
☐	✓ 0/224 B	IPv4 TCP	192.168.10.13	*	192.168.60.10	21 (FTP)	*	none		Allow Kali FTP to DMZ	Anchor Edit Copy Delete Toggle
☐	✓ 0/128 B	IPv4 TCP/UDP	192.168.10.13	*	WAN address	53 (DNS)	*	none			Anchor Edit Copy Delete Toggle
☐	✓ 0/0 B	IPv4 UDP	*	*	WAN address	*	*	none		Allow OpenVPN	Anchor Edit Copy Delete Toggle
☐	✓ 0/2 KiB	IPv4 TCP	*	*	192.168.60.10	80 (HTTP)	*	none		NAT WAN TO DMZ WEB SERVER	Anchor Edit Copy Delete Toggle
☐	✓ 0/6 KiB	IPv4 TCP	*	*	192.168.60.10	443 (HTTPS)	*	none		NAT WAN to DMZ HTTPS	Anchor Edit Copy Delete Toggle

[Add](#)
[Add](#)
[Delete](#)
[Toggle](#)
[Copy](#)
[Save](#)
[Separator](#)

7.6 VPN Client Configuration

A client configuration file with the .ovpn extension is used to connect remote users to the OpenVPN server.

The configuration contains the information required by the VPN client, including:

- VPN server address
- VPN port
- Protocol
- Certificates/keys

- Encryption settings
- Tunnel parameters

The client configuration is generated from the pfSense OpenVPN setup and can be imported into a compatible OpenVPN client.

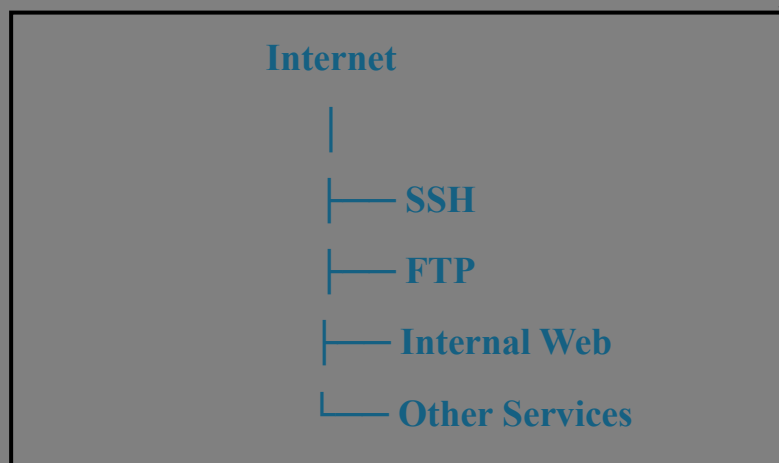
OpenVPN Client Export:

OpenVPN Clients		
User	Certificate Name	Export
alilhaider	internal-ca	- Inline Configurations: Most Clients Android OpenVPN Connect (iOS/Android) - Bundled Configurations: Archive Config File Only - Current Windows Installers (2.7.4-lx001): 64-bit 32-bit - Legacy Windows Installers (2.4.12-lx601): 10/2016/2019 7/8/8.1/2012r2 - Viscosity (Mac OS X and Windows): Viscosity Bundle Viscosity Inline Config
alikhawar	alikhawar	- Inline Configurations: Most Clients Android OpenVPN Connect (iOS/Android) - Bundled Configurations: Archive Config File Only - Current Windows Installers (2.7.4-lx001): 64-bit 32-bit - Legacy Windows Installers (2.4.12-lx601): 10/2016/2019 7/8/8.1/2012r2 - Viscosity (Mac OS X and Windows): Viscosity Bundle Viscosity Inline Config
shamir12	shamir12	- Inline Configurations: Most Clients Android OpenVPN Connect (iOS/Android) - Bundled Configurations: Archive Config File Only

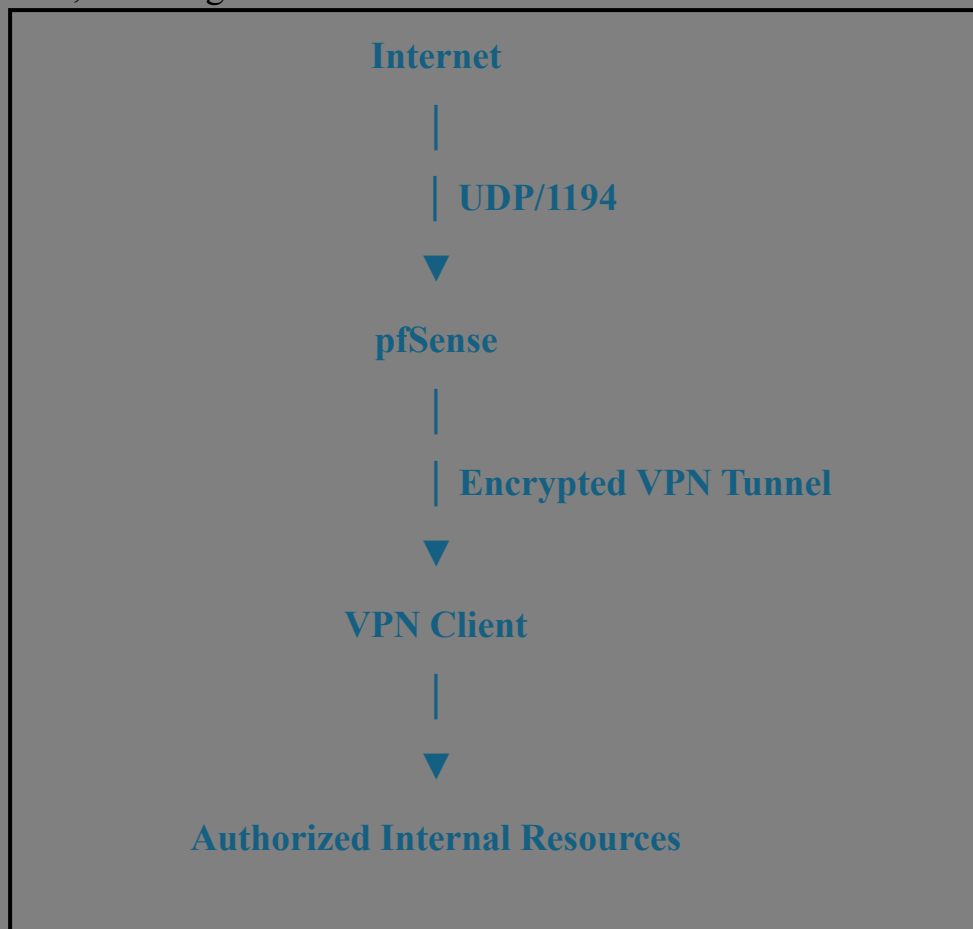
7.7 Remote Access Security:

The VPN provides a secure alternative to exposing internal services directly to the Internet.

Without a VPN, a remote user would need individual services to be exposed:



With OpenVPN, the design becomes:



This reduces the need to expose administrative services directly to the WAN.

7.8 VPN and Internal Network Access:

Once authenticated, a VPN client can be provided controlled access to the enterprise network based on pfSense firewall policies.

The VPN therefore acts as a controlled entry point:



pfSense Firewall



Access should remain limited to resources required by the authenticated user.

7.9 VPN Security Features

The OpenVPN implementation uses modern cryptographic options configured on pfSense.

The configured data ciphers include:

AES-256-GCM

AES-128-GCM

CHACHA20-POLY1305

AES-256-CBC

The configuration also uses: **SHA256**

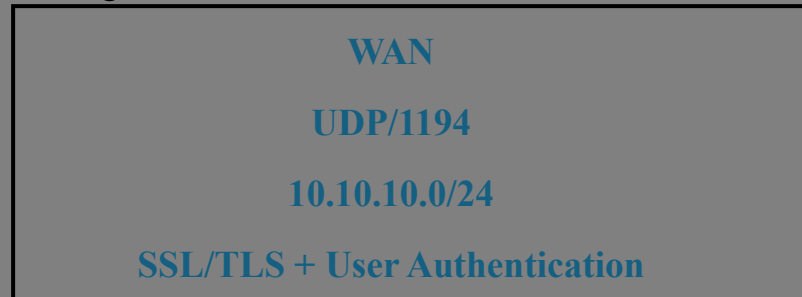
for message-digest functionality and 2048-bit Diffie-Hellman parameters.

These settings provide encrypted communication and secure key exchange for the laboratory VPN implementation

7.10 VPN Verification:

The VPN implementation was verified through the pfSense OpenVPN configuration and client setup.

The server was configured for:



Successful VPN operation indicates that the remote client can establish an encrypted tunnel to pfSense and receive an address from the VPN tunnel network.

OpenVPN Connection Verification

The screenshot shows the pfSense OpenVPN Status page. The top navigation bar includes links for System, Interfaces, Firewall, Services, VPN, Status, Diagnostics, and Help. The main content area is titled "Status / OpenVPN" and displays the following information:

ovpn1: internal-ca UDP4:1194 / Client Connections: 2

Common Name	Real Address	Virtual Address	Last Change	Bytes Sent	Bytes Received	Cipher	Actions
remoteuser alikhawar	192.168.10.8:43952	10.10.10.3	2026-08-13 01:39:44	4 KiB	6 KiB	AES-256-GCM	X +
internal-ca alishaider	192.168.10.4:65194	10.10.10.2	2026-08-13 01:37:31	5 KiB	9 KiB	AES-256-GCM	X +

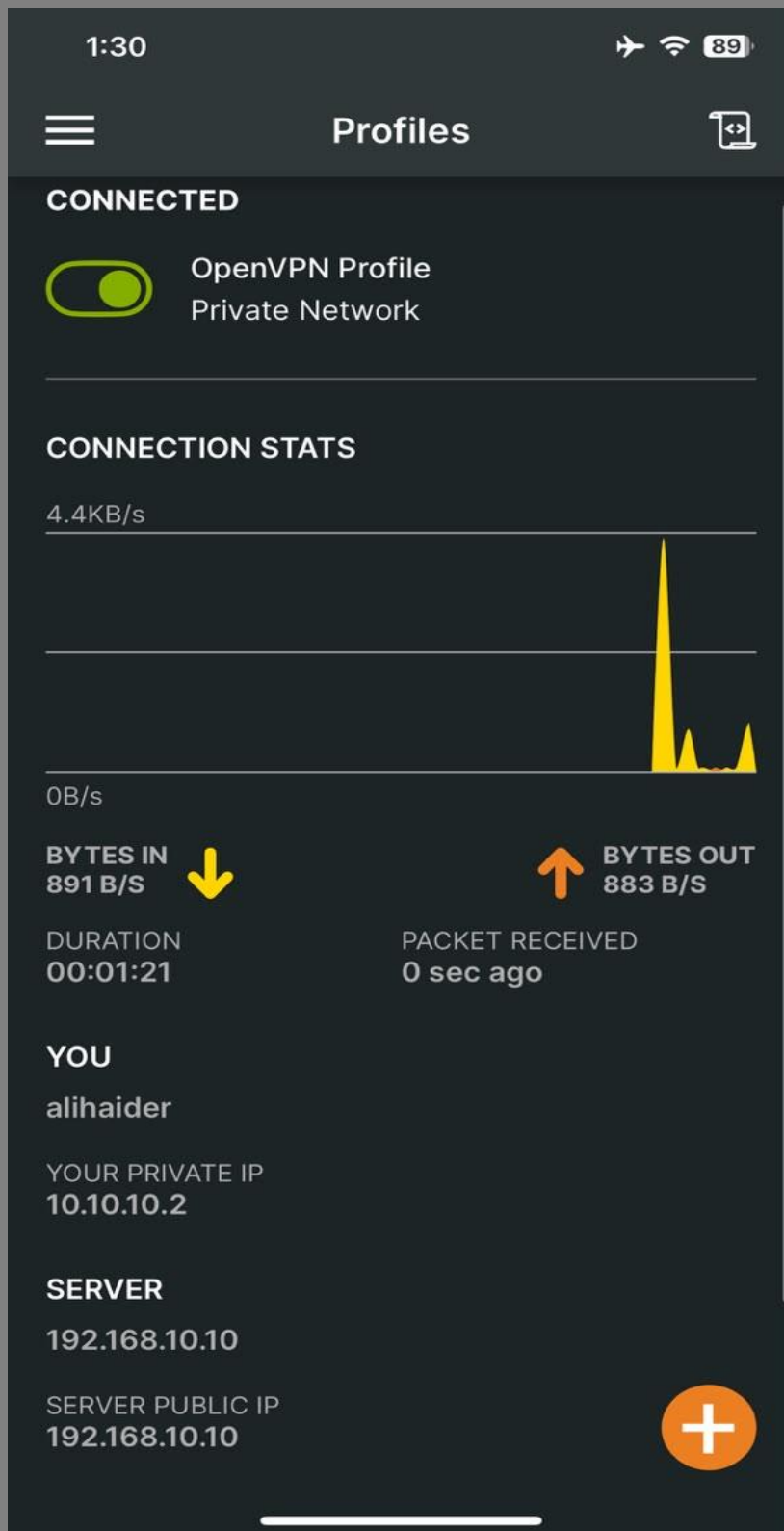
internal-ca UDP4:1194 Routing Table

Common Name	Real Address	Target Network	Last Used
remoteuser	192.168.10.8:43952	10.10.10.3	2026-08-13 01:39:46
internal-ca	192.168.10.4:65194	10.10.10.2	2026-08-13 01:39:01

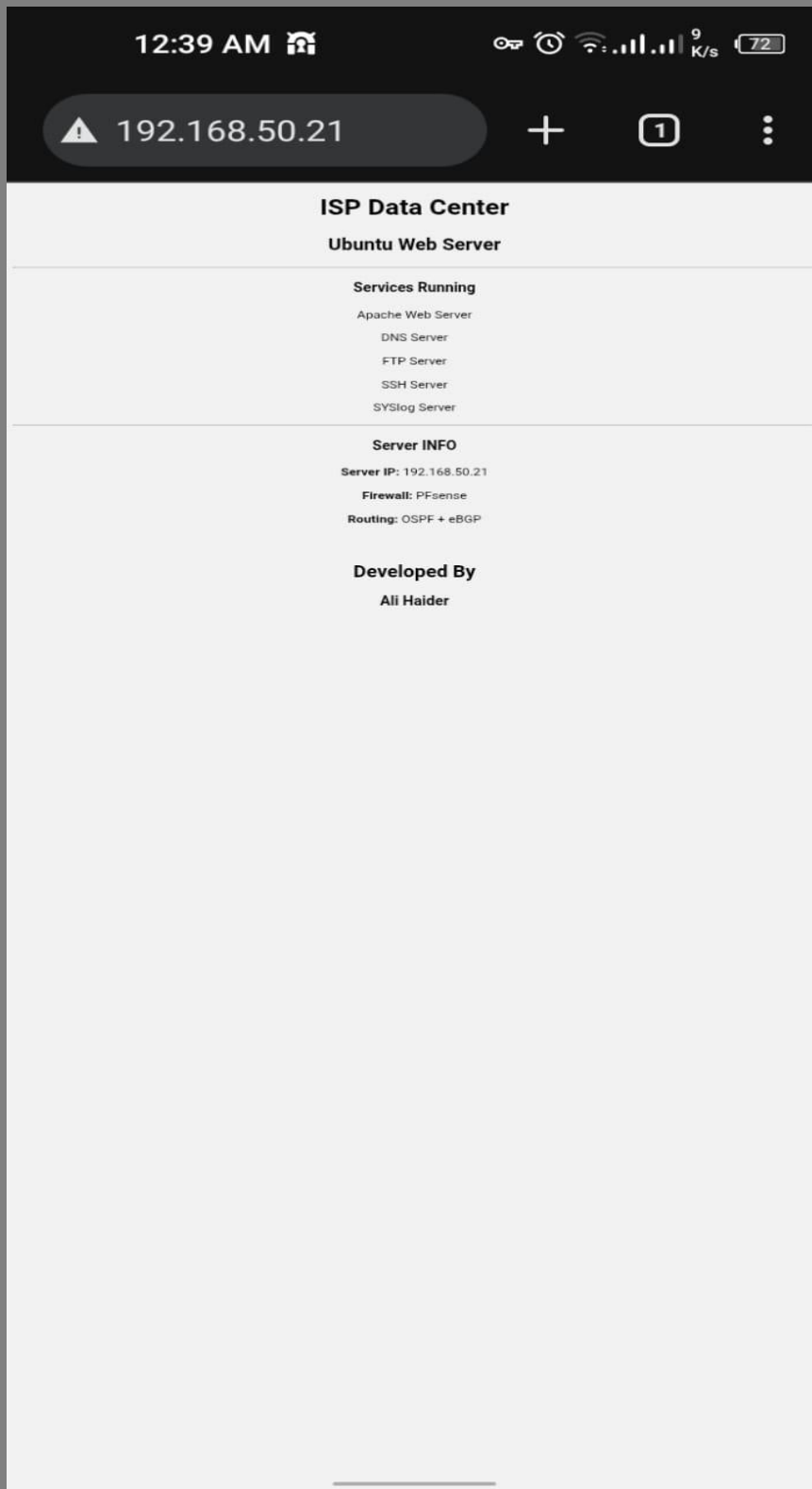
An IP address followed by C indicates a host currently connected through the VPN.

the OpenVPN client is successfully connected to the pfSense VPN server. This confirms that the VPN tunnel was established.

Successful OpenVPN Connection



OpenVPN client is successfully connected to the pfSense VPN server. This confirms that the VPN tunnel was established.



Access to Internal Web Server Through VPN

The web page is successfully accessed while connected through the VPN, demonstrating that the authenticated VPN user can reach the required internal network service secure

CHAPTER 10 — CONCLUSION AND FINAL RECOMMENDATIONS

10.1 Conclusion:

This project successfully designed and implemented a secure enterprise network infrastructure using GNS3, Cisco routing, pfSense, Ubuntu Server, Kali Linux, and OpenVPN.

The implementation combined multiple networking and cybersecurity technologies into a single working environment. OSPF provided dynamic internal routing, while eBGP enabled communication between different Autonomous Systems and allowed routing policies to be applied. HSRP provided gateway redundancy and improved network availability.

pfSense was successfully deployed as the central security gateway, providing firewall filtering, NAT, port forwarding, DNS integration, VPN access, and logging. A dedicated DMZ was implemented to host services that require controlled external access.

The Ubuntu Server successfully provided:

- Apache HTTP/HTTPS
- BIND9 DNS
- FTP
- SSH
- rsyslog

These services were tested from Kali Linux, and the required connectivity was successfully verified.

Security testing demonstrated that required services were accessible while unnecessary services were restricted. DMZ isolation was also verified by forcing traffic through the DMZ interface and confirming that access toward protected internal networks was blocked.

Overall, the project achieved its primary objective of creating a **functional, secure, and realistic enterprise network environment** while demonstrating practical networking and cybersecurity concepts.

10.2 Final Recommendations

Although the current implementation is functional, several improvements could make the environment closer to a production enterprise network.

Dedicated DMZ Server

A separate single-homed server should be deployed for the DMZ so that internal traffic cannot bypass the firewall through a second network interface.

VLAN Segmentation

VLANs should be implemented when an appropriate Layer-2 switch is available. Separate VLANs could be created for users, servers, management, and other security zones.

Centralized Security Monitoring

A SIEM platform such as Splunk or another security monitoring solution could be integrated to collect and analyze firewall, Syslog, and authentication events.

Trusted SSL Certificates

The laboratory currently uses a self-signed certificate for HTTPS. A trusted certificate authority should be used in a production environment.

Stronger Access Policies

Firewall and VPN rules can be further restricted according to user roles and business requirements, following the principle of least privilege.

10.4 Final Statement

The project provided practical experience in designing, configuring, troubleshooting, and securing a multi-layer enterprise network. It also demonstrated the importance of combining network availability, security controls, monitoring, and controlled access rather than relying on a single security mechanism.

The implemented environment can be further expanded in the future with VLANs, a dedicated DMZ server, centralized SIEM monitoring, stronger certificate management, and additional redundancy.

End of Report